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Review

A survey on UAV-assisted wireless communications: Recent advances and future trends☆

Xiaohui Gu, Guoan Zhang *

School of Information Science and Technology, Nantong University, Jiangsu, 226019, China

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ABSTRACT

Due to their characteristics of mobility, flexibility and adaptive altitude, unmanned aerial vehicles (UAVs), also known as drones, have immense potential applications in wireless systems. In particular, UAVs can operate as flying mobile terminals within a cellular network, and such cellular-connected UAVs have enabled several applications ranging from real-time video streaming to item delivery. Furthermore, the deployment of UAVs in wireless networks as flying base stations (BSs), servers or relays is expected to enhance network performance in terms of coverage, capacity, reliability and energy-efficiency. In this paper, a comprehensive tutorial on the potential applications and benefits of UAVs is presented. Specifically, UAVs as aerial BSs collecting data from ground sensors, transmitting power to energy-constrained devices and broadcasting information to ground users are explored along with representative design objectives. Then, UAVs as relay nodes for coverage extension, pathloss compensation and interference mitigation are surveyed comprehensively. Besides, UAVs as mobile servers performing edge computing for ground users are introduced, and various aerial offloading modes are reviewed thoroughly. Furthermore, radio and computation resource optimization, as well as mathematical tools and developed algorithms, are described in UAV-assisted communicating, relaying, and computing systems. Finally, open problems and potential research directions pertaining UAV-assisted communications are presented. In a nutshell, this tutorial provides key guidelines on how to design, analyze and optimize UAV-assisted wireless communication systems.

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* Corresponding author.
E-mail addresses: gu.xh@ntu.edu.cn (X. Gu), gzhang@ntu.edu.cn (G. Zhang).

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1. Introduction

To provide communications with increasingly stringent performance requirements in terms of latency, data rates and reliability, beyond fifth-generation (B5G) mobile communications aim to improve the user experience by offering consistent network performance throughout urban and rural environments. This is currently challenging since factors like radio coverage variations due to obstacles and terrain structures, high density and mobility of users significantly impact network capabilities. Without requiring costly infrastructure investments that typically accommodate a fixed amount of traffic, unmanned aerial vehicles (UAVs) can offer flexible radio and networking extensions [1]. UAVs can also hold control plane, orchestration and edge cloud capabilities on demand to enable scalable proximity services, offer network programmability and establish line-of-sight (LoS) communications. Hence, there is a growing industry and academic interest in integrating unmanned aerial platforms into the mobile network ecosystem.

1.1. Background

Due to their characteristics of agility, versatility, low cost and ease of deployment, UAVs have been rapidly applied in various domains, including real-time monitoring, providing communication coverage, remote sensing, searching and rescuing, delivery of goods, security and surveillance, precision agriculture and civil infrastructure inspection, etc. Many commercial UAVs on the market are equipped with WiFi connectivity so that they are remotely accessible. WiFi connectivity, however, may not be sufficient for beyond visual LoS communications needs, particularly for those requiring wide-area connectivity. Not surprisingly, the past few years have seen a surge of activities in utilizing established Long Term Evolution (LTE) networks for UAVs, and some field trials and study items are on the way [2–5].

Furthermore, with the commercialization of 5G (5th generation) communications and the prospect of 6G (6th generation), there have been significant researches in integrating UAVs into cellular communication systems for supporting reliable UAV commands and control, as well as mission-related payload communications [6]. Specifically, UAVs with their own missions could be connected to cellular networks as new aerial users. The advanced cellular technologies and almost ubiquitous accessibility of cellular networks also offer an effective option to strengthen legitimate UAV monitoring and management, and to achieve more robust UAV navigation by utilizing cellular signals as a complement to the Global Positioning System (GPS). Moreover, dedicated UAVs could be deployed as aerial base stations (BSs), access points (APs) or relays, to assist terrestrial wireless communications from the sky, leading to another paradigm known as UAV-assisted communications [7].

Recently, the increasing popularity of new resource-intensive applications, such as automatic driving, online gaming, health monitoring and virtual reality (VR), has led to a growing trend to execute the above attractive applications in our user equipments (UEs), such as mobile phones or handheld devices. However, this contradicts the size and battery capacity of portable UEs. Fortunately, mobile edge computing (MEC) [8] has been proposed, which pushes communication, computation and caching (3C) resources into the network edge. Besides, the future networks are expected not only to accommodate an unprecedentedly dynamic and heterogeneous environment but also to support on-demand hotspot areas and temporary activities in a fast and highly reliable manner. The traditional MEC system is inflexible and may suffer from high deployment costs [9], due to its fixed infrastructure. Hence, considering the flexible development of UAVs, deploying edge servers on them is considered a potential solution [10].

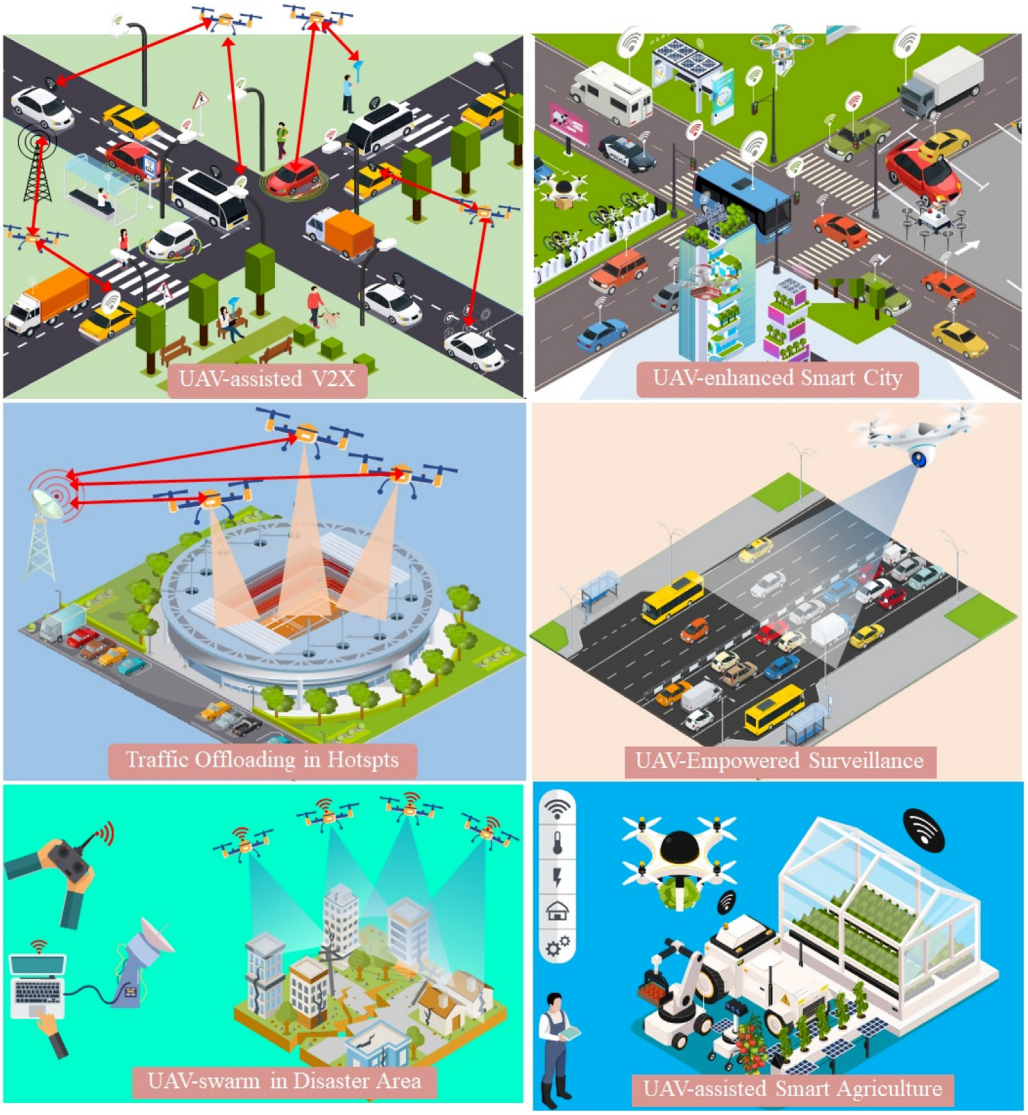


Fig. 1. Applications of UAVs in wireless networks.

In future wireless networks, as UAVs become more available, mobile users will not be restricted to terrestrial mobile stations. As shown in Fig. 1, there are plenty of applications of UAVs in wireless networks, such as UAV swarm networks for disasters [11], UAV-assisted vehicle-to-everything (V2X) communications [12], UAV-enabled smart city development, traffic offloading in hotspots, and UAV-aided surveillance and IoT networks. With the introduction of UAVs, the wireless network architecture will become a 3D structure, incorporating terrestrial and aerial network nodes, and the aerial-ground networks will be more dynamic than the fixed terrestrial communications network that we have today.

It is worth mentioning that the design/implementation of UAV-empowered wireless networks has a close relationship with the performance metrics, and some of popular metrics are listed as follows: *(1) Throughput/Capacity* acts as the tightest upper bound on the rate of information that can be reliably transmitted over a communication channel. It is a fundamental metric for spectrum efficiency, and capacity analysis of relay channels is a thriving area. *(2) Energy-efficiency (EE)* means the total information bits transmitted to UEs during the UAV flight time, which is normalized by the total UAV propulsion energy consumption during that time. How efficiently and optimally the UAV-enabled system can use onboard energy is a challenge because UAVs are fundamentally limited by a fixed energy supply. *(3) Outage probability (OP)* is an important performance metric for a wireless system. For slow-fading relaying channels, the communication capability could be evaluated by OP, that is, the probability that the transmission rate is not supported by the instantaneous channel capacity. *(4) Fairness* issue raised by multiple nodes with different abilities, should be carefully considered for UAV-enabled systems that require adaptive communications with dynamic resource allocations along with the UAV trajectory design, to enhance the communication performance. In other words, even though a certain user has a higher channel gain due to a shorter distance to the UAV and can support a higher data rate, one still reduces its allocated resources and reallocates them to worse users to maximize the minimum (max–min) average user rate and ensure fairness. *(5) Packet loss rate* is the average number of lost packets during one transmission period.

1.2. The roles of UAVs

For ever-increasing connected devices, it is possible to come across network congestion and long service latency, especially in hotspot areas or during temporary events such as the Olympic Games and NBA Games, and providing high-quality services in these high-density population areas with temporary high network traffic is important but challenging. Note that it is neither cost-effective nor realistic to deploy

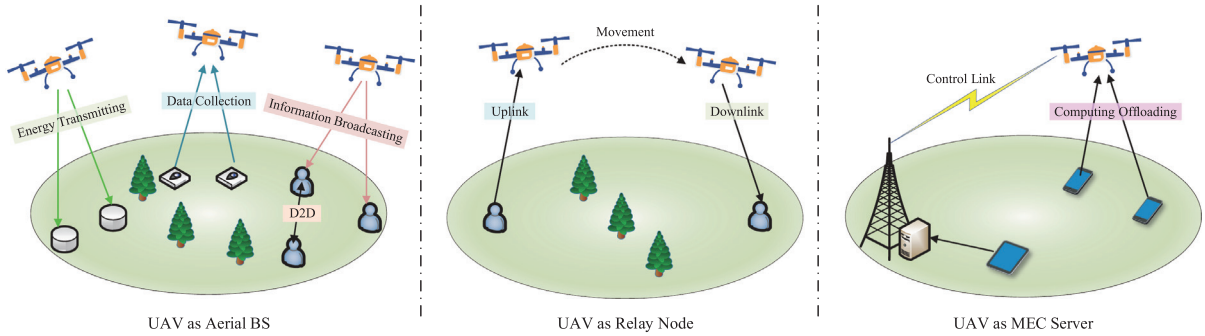


Fig. 2. The roles of UAVs.

many BSs for a small area or temporary events. In this regard, a BS or AP carried by a UAV (called a flying BS) is a promising solution to improve the network capacity. Due to easy deployment and flexible mobility, the UAV-carried BS and AP can also help extend network coverage for the terrestrial BSs having difficulty providing network service, especially in rural and disaster areas [13,14]. Additionally, the high altitude of UAV-carried flying BSs enables ground users (GUEs) to be easily distinguishable at different altitudes and elevation angles measured with respect to the UAV [7], which helps UAVs to effectively perform massive multi-input multi-output (MIMO), 3D network MIMO and mmWave communications. Furthermore, LoS paths of UAV-to-ground links ensure effective beamforming in both azimuth and elevation domains.

In the UAV-assisted wireless network, UAVs act in three key roles, as shown in Fig. 2. To be specific, UAVs with flexible mobility and LoS opportunities, are employed to support terrestrial networks in terms of data collection, information dissemination and power transfer. For instance, UAVs can be used as data collection devices that can gather vast amounts of data across various geographical areas within a city and deliver them to central cloud units for big data analytics purposes. In this regard, the combination of UAV and sensor technology facilitates timely and effective data collection, especially in complex, harsh, or remote environments. Besides, with their mobility and LoS opportunities, UAVs can support terrestrial networks for information dissemination and connectivity enhancement. In the case where device-to-device (D2D) performance is limited due to the short communication range of devices as well as potentially increasing interference, flying UAVs can facilitate rapid information dissemination by intelligently broadcasting common files among ground devices. On the other hand, mobile drones can introduce transmission diversity opportunities, thus boosting reliability and connectivity in D2D, ad hoc, and vehicle-to-vehicle (V2V) networks. Moreover, UAVs can be integrated into wireless sensor networks (WSNs) to minimize the challenges of sensor nodes' power consumption constraints, limited processing and storage capacities. UAV-enabled WSNs allow sensors to periodically establish connections rather than having a continuous connection with data centers; thus, the network throughput and nodes' lifetime can be improved. Additionally, integrating wireless power transfer (WPT) technologies into the UAV is prominent to prolong the lifetime of low-power rechargeable sensors/devices. Benefiting from the LoS links, the energy transfer efficiency can be greatly improved for the UAV-enabled WPT system, where the UAV is deployed as a mobile energy transmitter.

In the light of their computing capability, UAVs are used as relays to facilitate communications between two UEs or networks. UAVs acting as mobile relays embrace several added advantages compared to traditional ground relays. UAV relays are inexpensive and can be deployed quickly and easily when needed, making them ideal for emergencies and temporary events. Moreover, UAVs' high mobility provides an opportunity for enhancing network performance through location adjustment to facilitate LoS air–ground links. Particularly, multi-hop relay transmission technology can lay the foundation for aerial UAV relays.

In the 5G era, massive interconnections inevitably result in enormous volumes of data in communication networks, which imposes a rigorous challenge to the computation-limited and low-power UEs of Internet of things (IoTs). To deal with these issues, MEC is envisioned as a promising technology that can effectively alleviate the computational burden of UEs [15–17]. The amount of computing and caching resources in each MEC-mounted BS or AP is generally preset to allow UEs to offload partial or complete tasks to their associated BSs or APs for computing. While the resource demands from UEs are diverse and time-varying, underused or wasted resource issues remain in the big data scenarios supported by only MEC-mounted BSs, especially when there exists bursty traffic caused by some social activities or events. Besides, for these UEs located in remote or mountainous areas, they are unable to access reliable MEC service or even infrastructure coverage. Fortunately, mounting MEC servers in UAVs can help to address these problems by dispatching MEC-mounted UAVs to assist designated BSs or cover remote UEs [18].

Consequently, by integrating UAVs into terrestrial wireless networks, an air–ground integrated mobile network is formed, in which UAVs can either act as flying BSs and relays to improve communication services or work as aerial MEC servers to execute computation tasks offloaded from GUEs or the neighboring UAVs. Despite the promise of UAVs, we should address a number of technical challenges before using them for specific networking applications. For instance, when using a UAV as an aerial BS, key design considerations include the mission assignment, optimal 3D deployment, flight time and trajectory optimization and network planning. Meanwhile, in the UAV-relaying system, possible challenges include the working mode, handover management, channel modeling, relaying protocols, placement/trajectory and velocity/acceleration optimization of UAVs, security and reliability improvement and resources allocation. Moreover, in the edge computing scenario, performance characterization, multi-access management, low-latency control, 3D localization, workload balance, computation offloading decisions, user association and scheduling, wireless and computational resource allocation, and interference management are among the main challenges.

1.3. Related surveys

The applications of UAVs and UAV-ground communications have been extensively surveyed in plentiful literatures. However, existing works do not well reveal the guidance of UAV-assisted wireless networks. Table 1 compares the existing surveys and magazines, which indicates an imperative need for a comprehensive survey and tutorial on UAV-enabled applications and communications.

Specifically, from the perspective of UAV applications, the works in [19,20] presented the opportunities, requirements and challenges for UAV-empowered civil applications, [21] demonstrated how UAVs can be used for crowd surveillance based on face recognition, and [22]

Table 1
Comparison of related surveys and tutorials.

Category	Surveys	Focus point
UAV applications	[19], 2016	Civil applications of UAVs.
	[20], 2016	UAV networks for civil applications.
	[21], 2017	UAVs used for crowd surveillance.
	[22], 2017	UAVs for disaster management.
	[23], 2019	Multi-UAV enabled cyber–physical system (CPS) applications.
	[24], 2020	UAV-aided interference assessment for private 5G networks.
	[25], 2021	UAV-enabled ultra-reliable low-latency communications (URLLC) for 6G.
Channel modeling	[26], 2018	A2A and A2G channel modeling.
	[27], 2019	A2G channel modeling.
Routing protocols	[28], 2018	Cluster-based routing protocols for UAV networks.
	[29], 2019	Routing protocols for UAV networks.
UAV control	[30], 2018	Brain–computer interface (BCI) in designing control modules for UAVs.
	[31], 2018	UAV control and situational awareness under a jamming attack.
UAVs networks management	[32], 2016	Seamless and efficient inter-UAV connectivity.
	[33], 2018	Ultra-reliable UAV-swarm communications.
	[34], 2018	UAV orchestration framework for spectrum reusing and avoiding crowding.
	[35], 2019	Battery charging approaches to enable continuous wireless coverage of UAV.
	[36], 2019	Spectrum management for mmWave enabled UAV swarm networks.
	[37], 2020	A network slicing framework for UAV communications.
	[38], 2020	Coupling effects of communication, computation and control.
	[39], 2020	Enable UAVs to monitor UAVs.
	[40], 2021	Wireless UAV charging.
	[41], 2022	Topology management for UAV swarms.
	[42], 2022	RIS-assisted UAV systems.
UAVs cellular networks	[43], 2016	Beamforming for mmWave UAV cellular networks.
	[44], 2019	Integration of UAVs into cellular networks.
	[45], 2020	Integration synergies between 5G/B5G cellular systems and UAV technology.
	[46], 2021	mmWave beamforming enabled UAV communications and networking.
UAV-Assisted networks	[47], 2015	Multi-UAV aided vehicular network architecture.
	[48], 2016	UAV-enabled architecture for the delivery of aerial IoT services.
	[49], 2017	UAVs as relays between different public safety groups.
	[50], 2018	Power control in UAV-supported ultra dense networks.
	[51], 2019	UAVs as aerial BSs and cellular users in wireless networks.
	[52], 2020	Classification of UAVs applications with flight modes.
	[53], 2021	UAV-enabled WPT.
	[54], 2021	UAVs in 6G wireless communication networks.
	[55], 2022	Computation offloading in UAV-Enabled mobile edge computing.
	[56], 2022	Technologies associated with UAV applications.
This work		UAV-assisted communications: Benefits, applications (i.e., aerial BS, relay, MEC server), protocols, channel modeling, trajectory/position design, resources optimization, energy-efficiency, security and reliability, mathematical tools, open problems and opportunities.

envisioned UAVs assisting disaster management. For the cyber–physical security system (CPS) applications, the authors in [23] aimed to pinpoint the design challenges and potential methods of multi-UAV systems, [24] reported UAVs' potential use cases for capturing outdoor interference from private 5G systems, and [25] performed an overview of UAV-enabled URLLC networks.

In the perspective of UAV-related channel modeling, the authors in [26] provided a survey of air-to-air (A2A) and air-to-ground (A2G) channel modeling methods and categorized them as deterministic, stochastic and geometry-based methods, while [27] compared the characteristics, measurements and simulation results for the available A2G propagation models, followed by future research challenges in this domain. The cluster-, topology-, position- and social-network-based routing protocols, as well as the hierarchical, deterministic and stochastic routing protocols for UAV networks, were comprehensively surveyed and qualitatively compared in [28,29], accompanied by the discussion of design and implementation challenges. For the UAV control methods, as a promising candidate for deciphering brain signals into UAV control commands, the application of brain–computer interfaces in designing control modules for UAVs was surveyed in [30]; while in [31], a UAV-relay tactical data link was designed to guarantee reliable UAV control and situational awareness under a jamming attack.

Focusing on the management of UAV networks, the requirements from routing protocols peculiar to UAV networks, especially the needs for seamless handover and energy-efficiency, were discussed in [32,33] highlighted the ultra-reliable communication design for UAV swarms with a focus on the software protocol stack and radio frequency (RF) hardware. Motivated by separating the data plane from the control plane, [34] proposed a UAV orchestration framework that enables spectrum reusing and avoids crowding for A2A communications; [35] focused on UAV's battery lifetime, and conceived various charging approaches to enable continuous wireless coverage of UAVs; different spectrum management schemes were studied for the mmWave-enabled UAV swarm in [36]. A network-slicing-based framework for UAV communications was introduced in [37], where a certain amount of network resources were allocated according to diverse QoS requirements. The authors in [38] comprehensively revealed coupling effects of communication, computation and control on UAV networks. [39] described the idea of a dynamic UAV-aided radar network for detecting and tracking malicious UAVs. Various wireless charging techniques for UAV-enabled WPT were conducted by [40]. [41] reviewed a considerable number of topology control algorithms for UAV swarm networks. Taking promising potentials of the reconfigurable intelligent surface (RIS) technique, [42] focused on RIS-assisted UAV systems as well as UAV-assisted RIS systems. Considering mmWave UAV communications, the design of hierarchical beamforming codebooks, the Doppler effect as a result of UAV movement, the use of mmWave spatial-division multiple access and different ways of dealing with signal blockage were jointly investigated in [43]. [44] provided a comprehensive survey of the practical developments and standardization advancements that promote smooth integration of UAVs into cellular networks. [45] presented an exploration of integration synergies between 5G/B5G cellular

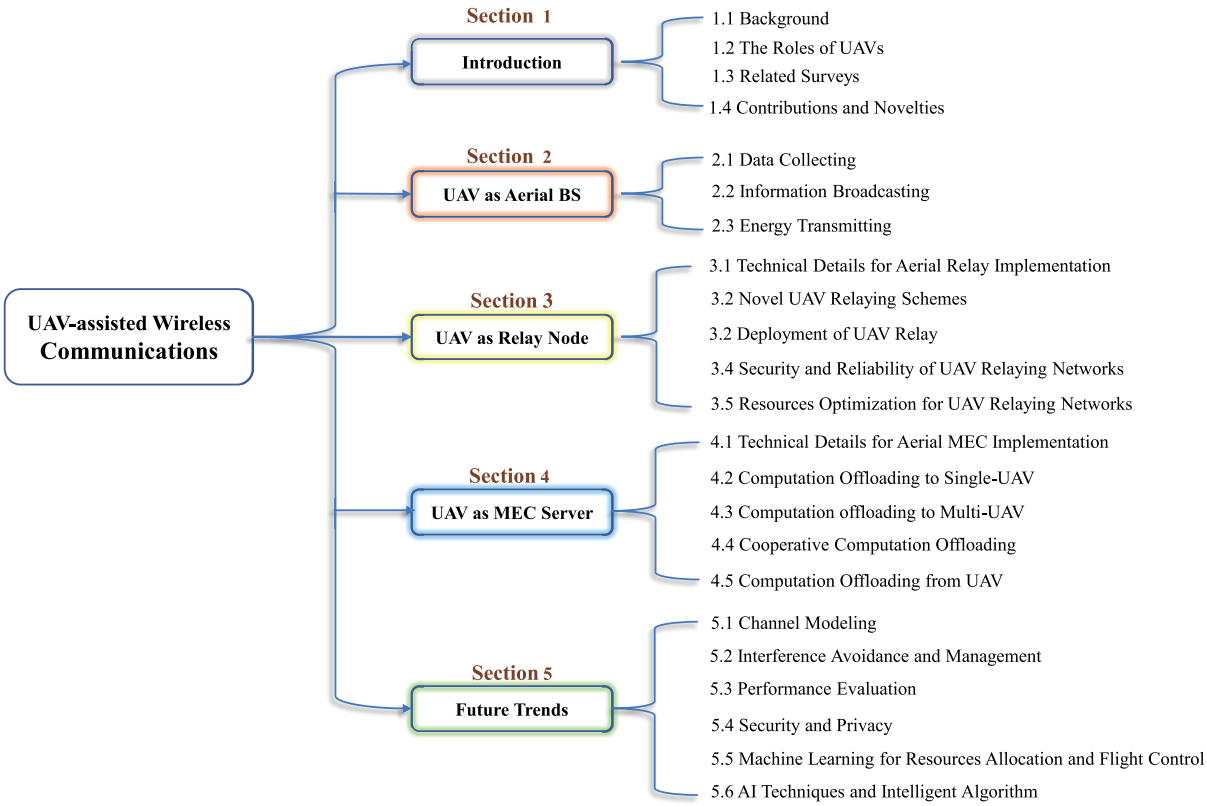


Fig. 3. Organization of this article.

systems and UAV technology. [46] surveyed mmWave beamforming-enabled UAV communications and networking in the perspective of physical layer technology.

In the region of UAV-assisted wireless networks, multi-UAV as intermediate relays were presented and discussed in vehicular networks [47] and different public safety groups [49]. In the condition of UAVs as aerial BSs, the technical and regulation issues associated with UAVs delivering IoT services were proposed in [48]. The applications and power control methods of UAV-supported ultra-dense networks were introduced in [50], the benefits and challenges of UAVs as aerial BSs and cellular users were surveyed in [51]. Specifically, the authors in [52] categorized the transmission models and corresponding application scenarios according to different flight modes of UAVs. [53] focused on single/multi-UAV enabled WPT and its applications in wireless communications and edge computing, while [54] introduced the role of UAVs 6G networks. [55] reviewed UAV-enabled MEC solutions in terms of computation offloading, and the authors in [56] focused on the supporting technologies that applied in UAV-enabled applications.

Although the above surveys address important UAV communication problems, they mainly limit their discussions to the methods of enhancing UAV communications or the performance of UAV-assisted systems from a communication perspective rather than the coverage of UAVs' various roles and the corresponding performance gains. Moreover, some surveys mainly focus on certain UAV applications, channel modeling, routing or control methods while overlooking broader networking problems. Clearly, the existing literatures on wireless networking using UAVs is largely fragmented, and given the rapid emergence of the topic in academia, industry and government, there is a clear need for a unified and comprehensive overview on how UAVs can be employed by wireless networks according to the network/users' requirements.

1.4. Contributions and novelties

The main contribution of this article is to provide a holistic and comprehensive tutorial on the roles that UAVs play in wireless communications and networking. To this end, the goal is to gather the state-of-the-art research contributions from the largely fragmented and sparse literature on UAV-assisted wireless networks and services. Moreover, this work emphasizes application scenarios, challenges, representative results, open problems, and analytical techniques in deploying UAVs as flying BSs providing data collection, energy supply and information broadcasting services; UAVs as mobile relays complementing limited communication capacity in wireless systems; and UAVs as aerial MEC servers broadening accessible and flexible computing resources for ground UEs. We also discuss the major opportunities and challenges that pave the way for the real-world deployment of UAVs as aerial communication and computation platforms. With the incessant growth in research revolving around the services relayed by UAVs, this article constitutes one of the comprehensive guides on how to fully take advantage of UAVs for wireless communications and networking.

The main body of this survey is comprised of two components: representative flight modes in UAV-assisted wireless networks and future directions for enhancing the performance of UAV-enabled wireless networks, as shown in Fig. 3. In Section 2, the UAVs operating as aerial BSs for collecting data from GSNs, broadcasting information, and providing energy for terrestrial UEs are respectively identified. In Section 3, when UAVs work as relay nodes, we first introduce the implementation issues, and then provide an overview of relaying schemes, UAV trajectory/placement design, security and reliability issues and resources management. In Section 4, after pointing out some important issues for UAV-enabled edge computing systems, the state-of-the-art research contributions in single-UAV, multi-UAV and cooperative offloading scenarios are comprehensively surveyed, and we evaluate the corresponding computation offloading and resources optimization schemes. Besides, the computation offloading from UAVs

Table 2
Acronyms used in this article.

Acronym	Text	Acronym	Text
UE	User equipment	GUE	Ground user equipment
SN	Sensor node	GSN	Ground sensor node
BS	Base station	AP	Access point
ET	Energy transmitter	ER	Energy receiver
2D	Two dimension	3D	dimension
EC	Energy consumption	EE	Energy efficiency
OP	Outage probability	SC	Secrecy capacity
WPT	Wireless power transfer	WIT	Wireless information transfer
DL	Downlink	UL	Uplink
AG	Air–ground	AA	Air–air
D2D	Device-to-device	DER	Decoding error probability
FD	Full-duplex	HD	Half-duplex
DF	Decode-and-forward	AF	Amplify-and-forward
TDD	Time division duplex	FDD	Frequency division duplex
Tx	Transmitter	Rx	Receiver
CSI	Channel state information	PLS	Physical-layer security
LoS	Line-of-sight	NLoS	Non-line-of-sight
RF	Radio frequency	EH	Energy harvesting
OMA	Orthogonal multi-access	NOMA	Non-orthogonal multi-access
TDMA	Time division multi-access	FDMA	Frequency division multi-access
BCD	Block coordinate descent	SCA	Successive convex approximation
AI	Artificial intelligence	ML	Machine learning
SNR	Signal-to-noise ratio	SINR	Signal-to-interference-plus-noise ratio
SISO	Single-input-single-output	SIMO	Single-input-multiple-output
MIMO	Multiple-input-multiple-output	MEC	Mobile edge computing

is also surveyed as a complementary study. In Section 5, we discuss some open issues and trends for future research. Section 6 concludes this survey. Some important acronyms used throughout this survey paper are given in Table 2.

2. UAV as aerial BS

Despite special aerial users, UAVs have acted as aerial communication platforms (aerial BSs), to complement terrestrial communications. This has inspired interesting use cases such as ubiquitous coverage, service recovery after natural disasters, information dissemination and data collection in IoTs and WSNs, and energy provision for GUEs. To realize a global vision of smart and connected communities and cities, UAVs can be used as data collection devices that can gather vast amounts of data across various geographical areas and deliver them to central cloud units for big data analytics purposes. On the other hand, UAV base stations can be used to simply enhance the coverage of wireless networks to enhance connectivity in information dissemination. Furthermore, UAV-enabled wireless energy transfer is a very promising solution for supporting D2D communications, machine-to-machine communications, and sensor nodes, and thus it helps prolong the network lifetime.

In this section, we discuss the key applications of UAV-mounted aerial base stations in order to paint a clear picture of how UAVs can indeed be used as flying wireless base stations. For each application of aerial BSs, we first present the outstanding benefits and advantages, and then we discuss the state of the art while also providing a comprehensive overview of recent results. For each application of aerial BSs, we provide the open challenges that should be pursued for practically deployment of UAVs as flying wireless BSs.

2.1. Data collection

By exploiting the maneuverability and short-range LoS opportunities, UAVs have been used to significantly enhance the connectivity and energy-efficiency of sensor networks. Since sensor devices are highly energy-constrained, they may be unable to transmit their data to a distant BS, especially during intermittent or poor coverage by terrestrial wireless infrastructures. Therefore, instead of installing dense small BSs, which are not flexible and economical to cover a large set of sensor devices, small UAVs are capable of collecting sensor data by moving towards the locations of sensor devices. This flexibility allows the transmission range of sensor nodes to be reduced to a minimal value, resulting in considerable energy savings and an increased network lifetime.

2.1.1. State of the art

For the implementation of aerial data collectors, this subsection comprehensively surveys the state-of-art literatures and lists them in Table 3.

In addition to the trajectory/placement design, the sleep and wake-up mechanism of sensors is another useful technique to save the energy consumption (EC) of GSNs in UAV-enabled WSNs. With such a mechanism, GSNs remain in the sleep state until they receive the waking up beacon signal with good strength from the nearby UAV, at which time they wake up and start sending data to the UAV, and return to the sleep state after transmission. While ensuring that the required amount of data is reliably collected from each GSN, Zhan et al. in [57] jointly optimized the GSNs' wakeup schedule and the UAV's trajectory to minimize the maximum energy consumption of all SNs. In contrast to [57] where the UAV needs to collect independent data from all GSNs, in [58], a UAV was employed to collect data from as many GSNs as possible for estimating an unknown parameter, and the UAV trajectory was also optimized in [58] to minimize the mean square error (MSE) for the estimation. To minimize the sum ECs of UAV and GSNs, the UAV deployment in terms of the number of stop points was optimized in [59], where a differential evolution algorithm with a variable population size was proposed to solve the formulated UAV's deployment problem. Considering the deadline-sensitive data, the UAV trajectory was optimized in [60] to accelerate the information collection, where the metric of age of information (AoI) [61] was applied to measure the freshness of sensed information. With the objective of maximizing the sum-data-rate of uploading information, the non-orthogonal multiple access (NOMA)-enabled UAV-assisted data collection system was investigated in [62,63]. Different from the conventional ground NOMA networks where the order of users' channel gains as well as the SIC decoding order are predetermined, both the orders cannot be predetermined since they are seriously coupled with the UAV's positions. In this regard, the authors in [63] exploited the signal-to-interference ratio (SIR) hypergraph and proposed a greedy coloring algorithm to find out the optimal NOMA groups. To practically capture the propagation environments of data offloading channels, the angle-dependent Rician fading was considered in [64,65], where the Rician

factors were determined by the corresponding UAV-GSN elevation angles. However, [64] is limited by LoS pathloss, and the probabilistic LoS channel model was adopted in [65]. Moreover, taking advantage of angle-dependent Rician fading, the authors in [64] approximated the outage probability (OP) requirement by a logistic function with respect to (w.r.t.) UAV 3D trajectory, and the UAV vertical trajectory was involved in [65] to balance the angle-distance trade-off for rate enhancement.

To further improve the efficiency of aerial data collection, Zhan et al. extended their work [57] to a multi-UAV-enabled WSN in [66], where multiple UAVs were dispatched to collect data from a set of ground SNs located in a given 2D area. The authors in [57] aimed to minimize the maximum mission completion time among all UAVs, while ensuring that each GSN can successfully upload a targeted amount of data with a given energy budget. In [67], they further studied the fundamental tradeoff between aerial cost (defined by the propulsion energy consumption and operation costs of all UAVs) and ground cost (defined as the EC of all GSNs). The multiple traveling salesman problem (MTSP) technique was used to design the UAV initial trajectory. Similar to [67] assuming that all UAVs share the same frequency to collect data with co-channel interference management, the work in [68] also optimized the workload and spectrum resource balance for UAVs. They evenly clustered IoT devices into several groups using the K-means method and assigned sub-channels to them based on the Matching Theory. For energy and spectrum efficiencies, Hattab et al. in [69] used the time-division duplexing (TDD) transmission protocol, which provides shared-spectrum access between cellular UEs' downlink communications and GSNs uplink data transmission to UAVs. Besides, the authors in [69] used stochastic geometry [70] to characterize the average available resources for UEs and GSNs, and modeled the locations of GSNs using an independent clustered homogeneous Poisson Point process (HPPP) [70].

Different from the aforementioned scenarios where the UAVs were employed to collect data from terrestrial sensors, some dedicated scenarios were also investigated by researchers more recently. For example, a short-endurance UAV was employed in [71] to collect data from buoy sensors, and the resonant beam charging was used to help the UAV get a better state of charge while ensuring reliable data transmission. The UAV can be implemented not only to assist terrestrial BSs in data collection but also to assist device positioning [72], to collect data over multiple GSNs in the presence of a jammer [73] and to simultaneously transfer energy and collect data to/from GSNs [74]. In addition, a linear topology scenario was investigated in [75], where GUEs are located in a straight line, and the UAV flies at a fixed altitude above the line connecting them, to reveal the fundamental rate limits of the UAV-enabled multiple access channel (i.e., NOMA, FDMA and TDMA). Moreover, Zhou et al. in [76] studied edge caching for multiple UAV-enabled radio access networks, in which UAVs serve the users based on the requested contents rather than the immediate close-by users.

Therefore, researchers have investigated various dedicated scenarios, such as employing short-endurance UAVs for data collection from buoy sensors, assisting device positioning, collecting data over multiple GSNs in the presence of a jammer, and simultaneously transferring energy and collecting data to/from GSNs. Moreover, edge caching for multiple UAV-enabled radio access networks has been studied, where UAVs serve users based on requested content rather than proximity.

2.1.2. Open challenge

While UAVs offer numerous advantages for data collection, realizing their full potential in sensor networks presents several challenges, including addressing data protection and privacy concerns, developing efficient data gathering algorithms, optimizing path planning based on mobile sensor node mobility, and ensuring data security and privacy. UAVs equipped with high-resolution and night-vision cameras can gather various types of information, raising concerns about the handling and usage of personal data. Compliance with data protection laws, such as the European Union's General Data Protection Regulation (GDPR) and the United Kingdom's Data Protection Act of 2018 [77,78], is crucial. These regulations aim to prevent the abuse of personal data and ensure that such information is managed safely and for specific purposes. Consequently, it is vital to carefully manage UAV-powered data collection to protect the security and privacy of personal data while adhering to relevant regulatory frameworks. Developing robust and efficient data gathering algorithms is necessary to facilitate autonomous collection and processing of data. These algorithms should enable UAVs to dynamically adjust their flight paths and communication strategies based on the sensor network's state and environmental factors. Investigating the mobility of mobile sensor nodes helps in designing fast and energy-efficient path planning for UAVs, allowing for data collection from an initial point to the destination in the shortest amount of time with minimum energy consumption. Ensuring data security and privacy is vital to protect against unauthorized access, data leaks, or malicious attacks in UAV-enabled data collection that may involve sensitive or private information.

2.2. Information broadcasting

UAV-enabled information dissemination or broadcasting is an important use case of UAV-enabled communication systems [7], with various application scenarios such as for public safety and emergency responses [79], video streaming [80], military operations and intelligent transportation systems [81,82]. Different from conventional broadcasting with static transmitters (e.g., terrestrial BSs), where the broadcasting performance is fundamentally limited by the user's being far from the transmitter, UAV-enabled broadcasting can overcome this issue by exploiting its high mobility through adaptive trajectory design.

2.2.1. State of the art

In recent years, many researchers have contributed to the implementation of UAV-assisted information broadcasting, and some relevant works are summarized in Table 4.

In the UAV-assisted information broadcasting system, Wu et al. in [83] considered a UAV flying at a constant altitude to send independent information to two GUEs at different fixed locations. With the aim of characterizing the capacity region of the considered system, by jointly optimizing the UAV's trajectory and transmit power/rate allocation over time under UAV's maximum speed and maximum transmit power constraints, they have concluded some essential insights into UAV trajectory design. Specifically, given the speed and flight duration of the UAV, [83] shows that the optimal UAV trajectory should follow the hover-fly-hover (HFH) structure, i.e., the UAV, at its maximum speed, flies unidirectionally between a pair of initial and final points, which are above the ground users. [83] respectively studied and compared the capacity gains obtained by superposition coding (SC)-based non-orthogonal transmission and TDMA-based transmission, and the OFDMA transmission was investigated in [84]. Considering the user communication delay requirements, the minimum-rate ratio (MRR) constraint was introduced by [84], which was defined as the minimum instantaneous rate required over the average achievable throughput, and it also reveals the tradeoffs among the system max–min throughput and users' communication delay and MRR constraints. Through jointly optimizing the UAV's altitude, transmit beamwidth and power and available bandwidth for each user, the max–min rate optimization problem was studied in [85], where the UAV needs to pair two NOMA users. With a similar objective, a UAV in [86] was deployed at the center of a virtual cell to transfer information to GUEs while sharing the same spectrum with D2D pairs. The UAV in [87] combines TDMA and FDMA to communicate with multiple users, with the goal of maximizing energy-efficiency defined as the ratio of user throughput to the UAV's energy consumption.

Instead of focusing on improving the *energy-efficiency*, the energy-constrained *completion time* minimization was investigated in [88,89],

Table 3
Comparison of literatures about UAV-empowered data collection.

Scenario	Ref.	Objective	Optimization items	Main Contributions
An UAV collects data from multiple GSNs	[57]	Min–Max the ECs of all GSNs	GSNs' wake-up schedule and UAV's trajectory	Each GSN employs the sleep/wakeup scheme.
	[58]	Max no. of visited GSNs	UAV 2D trajectory	Applicable to time critical and delay tolerant sensing measurements
	[59]	Min sum-EC of UAVs and GSNs	UAV 2D deployment	Optimize the number and locations of stop points simultaneously for the UAV's deployment
	[60]	Min no. of expired packets	UAV 2D trajectory	Combine the AoI characteristic, the deadline constraints and artificial intelligence together.
	[62]	Max sum-rate of GSNs	UAV 2D deployment, GSNs power control	Users' channel gains and SIC decoding order are coupled with UAV position.
	[63]	Max sum-rate of SNs	UAV 2D placement, GSNs clustering and power control	The SIR hypergraph and the greedy coloring algorithm was employed to optimize NOMA groups.
	[64]	Max–Min average rate of GSNs	UAV 3D trajectory, GSNs scheduling	OP requirement is approximated by a logistic function w.r.t. UAV 3D trajectory.
	[65]	Max–Min average rate of GSNs	UAV 3D trajectory, GSNs scheduling	Hybrid offline-online optimization method using statistical and real-time CSI.
Multi-UAV collect data from multiple GSNs	[66]	Min–Max mission completion time	UAV 2D trajectory, GSNs wake-up scheduling and association	GSN employs sleep/wakeup scheme; Tradeoff between the energy budget of GSNs and the mission completion time of UAVs.
	[67]	Min weighted-sum of aerial and ground costs	UAV trajectory, GSNs wake-up time allocation and power control	UAVs collect data at the same frequency with co-channel interference management.
	[68]	Min sum-transmit powers of GSNs	UAV 3D placement, bandwidth allocation, GSNs power control	Multiple GSNs are divided into several groups with the same number to balance the workload of UAVs.
	[69]	Max average energy-efficiency of GSNs	GSNs power control	A shared-spectrum access between massive GSNs and cellular UEs; Model GSNs' locations with an independent clustered HPPP process.
A short-endurance UAV collects data from buoy sensors	[71]	Max power transmission efficiency	UAV 2D trajectory, charging station's power control	Employ resonant beam charging to help UAV get a better state of charge while ensuring reliable data transmission.
An UAV as both a data collector and an aerial anchor	[72]	Min–Max sum-EC of GSNs	UAV 2D trajectory, GSNs transmission scheduling	UAV assists terrestrial BSs in data collection and device positioning; Consider the uncertainty of device location for problem formulation.
An UAV collects data over multiple GSNs in the presence of a jammer	[73]	Max–Min expected rate of GSNs	UAV horizontal and vertical trajectories, GSNs transmission scheduling	The UAV sequentially collects data while mitigating the ground jammer.
An UAV transfers energy and collects data to/from GSNs	[74]	Min the average AoI	UAV 2D trajectory, time allocation for EH and data collection	The UAV acts as both a energy source and data collector; Nonlinear EH model.
GSNs in a straight line communicate with a UAV	[75]	Max the average sum-rate of GSNs	UAV 1D trajectory, bandwidth/time allocation	NOMA, FDMA, TDMA transmission protocols are respectively investigated.
Multi-UAV act as edge caching	[76]	Max spectrum efficiency	Popularity threshold of contents	UAVs serve GUEs based on the requested contents.

revealing that the throughput becomes maximum via transmit power optimization and the completion time can be reduced via UAV trajectory optimization. Similar to the conclusion in [83], it has been shown in [89] that the optimal UAV trajectory only needs to constitute connected line segments, which can be simplified by finding the optimal set of waypoints and the optimal speed along the path connecting those waypoints. In [90], the completion time and energy consumption minimization were respectively evaluated for UAV-enabled multicasting, where the UAV sweeps the destination area with a zigzag pattern when disseminating some common information to GUEs. Different from most works assuming the quasi-static channel, the time-varying A2G channel was modeled in [90] to design the UAV's beamwidth.

In addition, to overcome the challenge of the limited endurance of UAVs, the content-centric communication scheme was investigated in [91], by taking advantage of proactive caching at the users. In [91], the UAV proactively transmits files to a subset of selected ground nodes that cooperatively cache all the files, and then each node can acquire the desired file either directly from its local cache or from its nearest neighbor via D2D communications. In [92,93], a UAV was deployed to a malfunction area for recovering communications. Instead of focusing on the UAV path planning to serve as many users as possible, a user-centric cooperative scheme was proposed in [92], where a UE can choose to be served by the UAV, the nearest ground BS, or both the UAV and the nearest ground BS, according to the quality of communication connections.

The research observation in [94] shows that, compared to the *single-UAV* case, the throughput and access-delay tradeoff can be significantly improved by the use of *multi-UAV*, since the users could be served in parallel in the multi-UAV enabled network. The energy-efficiency of a similar definition in [87], was investigated in [95] for multi-UAV-enabled multi-cell networks, where GUEs in each cell access the UAV with OFDMA protocol. After the proof of Nash equilibrium (NE) in the non-cooperative game, where each cell selfishly tries to maximize its own energy-efficiency without exchanging information or decisions with other cells, the authors in [95] further extended it to deal with dynamic time-varying situations where the UAVs can dynamically update their scheduling and resource management strategies. [96–98] focused on the DL transmission in a heterogeneous network, with the coexistence of one terrestrial BS and multiple UAVs as aerial BSs. Specifically, the user association problem was formulated as a coalition game in [96] considering the issues of load balancing and user fairness. The multi-UAV enabled wireless mesh network with 2D/3D homogeneous PPP distribution was modeled in [97], and the impacts of UAV's height,

transmit power, density and vertical range on the coverage probability were also analyzed, followed by the analysis of performance improvements brought by spectrum sharing. The coverage of the ground BS was split into non-overlapping sub-sectors in [98], and each of which is served by a UAV for DL transmissions. The cooperation between the BS and UAVs in [98] was evaluated during one timeslot, in which the ground BS transmits NOMA messages to multiple UAVs, and the UAVs use successive interference cancellation (SIC) to decode the received messages. In the subsequent timeslot, all the UAVs simultaneously transmit the decoded data to their destinations in an FDMA manner.

By applying stochastic geometry, the coverage probability was analyzed in [92,97], which is defined as the probability of the event that the signal-to-noise ratio (SNR) [92] or SINR [97] is higher than a threshold for successful communication. Directional antennas were applied in A2A and A2G links to reduce interference in [97], and the performance improvement using directional antennas was compared with that using omnidirectional antennas. The design of transmit beamforming at the ground BS and UAVs were respectively optimized in [92,98] using optimization theory.

2.2.2. Open challenge

The main challenge of UAV-enabled information broadcasting lies in addressing the limitations and inefficiencies in existing approaches to channel state information (CSI) estimation in 5G massive MIMO systems. Current methods depend on algorithm design and system optimization, leading to significant transmission overheads, bandwidth waste, and energy consumption. Specifically, UAVs assisting BSs in downlink broadcasting transmission face challenges due to their high-mobility, three-dimensional operations, and inter-cell interference, requiring more frequent CSI estimation compared to GUEs. Additionally, UAVs are constrained by energy, size, computation power, and memory, with resource depletion potentially causing severe consequences such as broadcasting service disruption or UAV crashes. To overcome the challenges and inefficiencies in UAV-enabled information broadcasting, it is essential to develop innovative CSI estimation methods for 5G massive MIMO systems that minimize transmission overheads, reduce bandwidth waste, and optimize energy consumption while accounting for the unique constraints and operational characteristics of UAVs.

2.3. Energy transmitting

With the ambitious goal of connecting everything in the world via wireless communication technology, a mass of IoT devices is distributed in the given region randomly or determinedly to sense and gather information seamlessly. However, the power supply for energy-constrained IoT devices is always a tricky problem for designing reliable and robust networks, and it is desirable to provide a controllable and sustainable power supply mechanism for these IoT devices. UAV-assisted wireless power transfer (WPT) has drawn significant research interest in terms of enhancing the efficiency of WPT for charging low-power smart devices. Typical applications include smart cities, smart factories, maritime communications, wireless sensor networks and IoT where wired charging is unavailable. By exploiting their controllable mobility, UAVs mounted with energy transmitters can properly adjust their trajectory over time to reduce their distances from target smart ground devices, thus improving the efficiency of WPT.

2.3.1. State of the art

So far, many works have been done on wireless-powered communication networks (WPCNs), where the nodes first harvest the RF energy transferred by the energy source (ES) in the DL and then conduct wireless information transmission (WIT) by using the harvested energy. According to whether the energy transmission and data receiving are conducted by the same device, WPCNs can be generally classified into two types: (1) the WPT network, where the ES only transmits RF energy to nodes, and (2) the WPCN, where the ES not only transmits RF energy to nodes but also receives data transmitted by nodes [99]. In the following, we survey the works about UAV-enabled WPT, following by the works related to UAV-enabled simultaneous WIT and WPT, as shown in Table 5.

In practice, the UAV flight duration is finite due to the limitation of onboard battery capacity, and hence, maximizing the energy-efficiency of UAVs has been widely studied in existing works. Specifically, after investigating the optimization of timeslot scheduling for each UE and durations allocation for charging, the research observation in [100] shows that the efficiency of energy harvesting (EH) is affected by the charging sequence and pathloss model. As the trajectory/placement design for UAV significantly affects the charging efficiency, many works, i.e. [101–105], have focused on the optimization of UAV's position/trajectory. Specifically, with the assumption of LoS and static A2G channels, [101] considered two charging schemes for both 1D and 2D WPT systems: Scheme 1, the UAV charges those sensors only when it arrives and hovers at their top; Scheme 2, the UAV starts charging ground nodes when it is flying closer to them; what is more, Scheme 2 shows a better energy-efficiency than Scheme 1. Assuming that the UAV transmits energy when hovering and that A2G channels are time-varying, the authors in [102,103] have concluded that the UAV should successively hover at a set of fixed points for an optimized duration, and the UAV should fly among these hovering points at the maximum speed. Additionally, the upper bound of the number of hovering points was mathematically derived by the authors in [103], and for effectively addressing the trajectory design problems, they also designed a differentiation approach for the approximation of the integral function, which has high potential for the trajectory design. Considering the practical scenario where the UAV has partial prior knowledge of energy receivers' distribution, the UAV in [104] adopts the radio map technique to efficiently acquire detailed geographical channel information. Observing from the analog beamforming design for multiple RF chains, numerical results in [105] illustrate that the received power can be constant when the UAV flies below a minimum altitude. Instead of maximizing the volume of harvested energy, a UAV was employed in [106] to help improve the throughputs of D2D pairs by providing them with RF energy.

Moreover, some published works have intended to investigate the combination of WPT and WIT in UAV-enabled WPCNs. For example, [107] enabled a UAV to charge users when the UAV is in hovering and flying states and collect information only when the UAV is hovering above each of the GUEs. The objective of [107] is to maximize the minimal throughput among ULs through the successive hovering-flying-hovering trajectory design and resource allocation for DLs and ULs. A UAV-enabled mmWave WPCN was investigated in [108], where a UAV as a constant power supplier coordinates WPT/WIT to/from GUEs, to maximize the sum-rate of ULs, which was analytically expressed as a function of the UAV's position. To minimize the transmission completion time for collecting a given number of bits per node, [109] jointly optimized the UAV's locations for conducting RF energy transfer and data receiving, the energy transfer duration at each location, and the nodes' transmit powers and durations. Rather than focusing on the case where the UAV performs WPT and WIT separately, an FD UAV was employed in [110] for simultaneous information collection and WPT. In such a case, the UAV broadcasts energy when flying and hovering and collects information only when hovering. As a result, the time allocation for hovering and flying has different trends for different goals, i.e., the sum-throughput maximization, total-time minimization, and sum-energy consumption minimization. Specifically, the superiority of implementing UAVs in WPCNs was verified in [107,109], compared with the employment of conventional APs located in optimal positions.

Table 4
Comparison of literatures about UAV-empowered information broadcasting.

Ref.	Objective	Optimization items	Features	Conclusions
[83]	Max achievable sum-rate	2D trajectory, transmit power/rate allocations	Respectively investigated two special cases with asymptotically large/low UAV flight duration/speed.	The optimal UAV trajectory should follow a general hover-fly-hover structure and UAV flies at the maximum speed.
[84]	Max–Min average rate	2D trajectory, bandwidth allocation, power control	Introducing a minimum-rate ratio constraint defined as the minimum rate required over the total achievable average throughput.	Different MRR setups affect the tradeoff between the system throughput and the user communication delay.
[85]	Max–Min achievable rate	UAV's altitude, transmit beamwidth and power, bandwidth allocation	The UAV can employ NOMA to pair each near user with one of the far users.	A clear rate gain by jointly optimizing all the parameters rather than optimizing only a subset of parameters.
[86]	Max–Min achievable rate	Bandwidth allocation, D2D assignment, power allocation for D2D pairs and UAV	D2D pairs share spectrum with UAV-enabled communications.	The cognitive network with spectrum sharing outperforms the non-cognitive network using a partial bandwidth for D2D users only.
[87]	Max energy-efficiency	3D trajectory, UEs scheduling and association, power and bandwidth allocation	Establish an energy-efficiency model for UAV 3D flight by analyzing the energy consumption of UAV and the user throughput.	Maximizing energy-efficiency is more efficient than only maximizing the throughput.
[88]	Min mission completion time	2D trajectory, UAV transmit power	Improve the algorithm by maximizing the throughput instead of minimizing the completion time.	The throughput becomes maximum via the transmit power optimization and the completion time can be reduced via the trajectory optimization.
[89]	Min mission completion time	2D trajectory	Ensuring each UE is able to successfully recover the file with a targeting probability.	Compared to static transmitter, UAV-enabled multicasting with optimized trajectory achieves significant performance gains.
[90]	Min mission completion time and UAV's EC	UAV speed, altitude, and antenna beamwidth	UEs' locations are unknown at UAV.	The optimal altitude that minimizes the completion time is always the maximum allowed.
[91]	Min weighted-sum costs	2D trajectory, file caching policy, UEs scheduling	The UAV pro-actively transmits the files to a subset of selected GNs, to be retrieved by each GUE.	Joint caching and trajectory optimization achieves significant performance gains in terms of file caching and retrieval costs.
[92]	Max coverage probability	Cooperation scheme	A UE is served by the UAV only, the nearest BS only, or both the UAV and BS.	An expression for the coverage probability achieved by a typical UE is obtained and verified.
[93]	Max served UEs	Path planing	Unknown user distribution and traffic demand; Divide the entire area into multiple equal grids.	The proposed online learning algorithms outperform the baseline helical path in terms of the total number of served users.
[94]	Max–Min average rate	2D trajectory, UAV power control, UEs scheduling and association	Multiple UAVs share the same spectrum and each UAV serves its associated GUEs via TDMA.	Compared to the single-UAV case, the throughput-access delay tradeoff is shown to be significantly improved by the use of multiple UAVs.
[95]	Max energy-efficiency	3D placement, UAV power control, UEs association	UAVs coordinated and uncoordinated approaches.	The performance of the coordinated approach outperforms than that of the uncoordinated approach under both dynamic and stationary scenarios.
[96]	Max network utility	UEs association, power allocation for DLs	Develop a coalition matching game approach to balance the traffic among the UAVs and the terrestrial BS.	The achieved rate obtained by the collation matching game is higher than that obtained by the maximal SINR-based scheme.
[97]	Max coverage probability	UAVs' height, transmit power, density and the vertical range	2D/3D homogeneous PPP modeling for the distribution of UAVs; Spectrum sharing between BSs and UAVs.	Compared with omnidirectional transmission, the performance of spectrum sharing with directional transmission is better.
[98]	Max achievable sum-rate	2D placement, radio resource allocation, decoding order	The coverage is splitted to non-overlapped sub-sectors and each of which is served by a UAV.	The cooperative NOMA scheme between UAVs and BS is affected by the change of UAV's position.

2.3.2. Open challenge

Most of the aforementioned works conceive of a single UAV transmitting energy for multiple users, and multiple UAVs serving a large number of users over a large area needs further research. In this condition, collaborative downlink energy beamforming and efficient uplink interference control via multi-UAV trajectory design can be employed to enhance the network performance. Besides, the above works assumed a fixed mission period for UAVs. However, in practice, the UAV mission period should be set according to various factors, such as various latency requirements and different battery sizes of energy receivers, as well as the battery lifetime and power consumption of the employed UAVs. Thus, how to design the UAV mission period under these practical considerations is an interesting problem worth further study. On the other hand, the linear EH model is too general for smart devices, which pushes the investigation into efficient rectification of the power harvester and remaining battery energy.

3. UAV as relay node

UAVs, as mobile relays, are characterized by a number of additional advantages over conventional ground relays. UAV-assisted relays are ideal for emergency and temporary events since they are affordable and can be deployed quickly and efficiently as needed. Furthermore, the high mobility of UAVs poses an opportunity to improve network performance through customization to the environment.

Table 5
Comparison of literatures about UAV-empowered energy transmitting.

Issue	Ref.	AG channel	Objective	Features	Contributions/Conclusions
WPT	[100]	Static	Min the total time required to charge field ERs	EH efficiency is evaluated by shadowing statistics of pathloss and non-linear RF-to-direct current conversion efficiency.	The number of field nodes that can be supported is affected by the charging sequence.
	[101]	Static	Max the total harvested energy	Two charging schemes: the UAV only charges ERs when it arrives and hovers at their top; the UAV starts charging while flying close to ERs.	Hovering-and-flying-EH shows a better energy-efficiency than hovering-EH.
	[102]	Time-varying	Max the total harvested energy	Both sum-energy and min-energy maximization problems were investigated.	The UAV should hover over an optimal set of fixed locations, with optimal hovering time allocations among them.
	[103]	Time-varying	Max–Min the harvested energy among all ERs	The UAV successively hovers at a set of points each for an optimized duration, and flies among these hovering points at the maximum speed.	The upper bound of the number of hovering points is derived; The proposed differentiation approach for the approximation of an integral function, is of high potential in addressing trajectory design problems.
	[104]	Time-varying	Max–Min the received energy among all ERs	The UAV adopts the radio map technique to efficiently acquire the detailed geographical channel information; The UAV partially knows ERs' locations subject to norm-bounded errors.	Proposed radio-map-based design superiors to benchmark designs under LoS and probabilistic LoS channels.
	[105]	Static	Max the total harvested energy	Analog beamforming for multiple RF chains.	The received power is constant when UAV flies up to a minimum altitude, but drops afterwards.
	[106]	Static	Max sum-throughput of D2D pairs	Management for multiple D2D pairs which share the same spectrum.	Benefiting from higher spectrum utilization, the average throughput grows with increasing number of D2D pairs.
WPT & WIT	[107]	Time-varying	Max–Min UL throughput	The UAV successively hovers over a set of fixed locations for WPT and WIT.	The trajectory design of UAV outperforms the conventional scheme with a static AP located at the optimal location.
	[108]	Static	Max the throughput of all ground nodes	Tailor the sequential unconstrained convex minimization based algorithm for UAV enabled wireless powered communication networks.	The proposed sequential unconstrained convex minimization based algorithm obtains the globally optimal solution.
	[109]	Static	Min the transmission completion time	UAV positions and time durations for WPT and WIT were jointly investigated; The UAV hovers above each of the ground nodes for uplink WIT.	The transmission completion time achieved by using the UAV is greatly smaller than that achieved by using the fixed AP.
	[110]	Time-varying	Max sum-throughput, Min total-time, Min UAV's EC	The UAV has two antennas to enable simultaneous WPT and WIT; The UAV performs WPT when flying and hovering, and performs WIT when hovering.	The time allocation between hovering and flying time has different trends for different goals.

For a better understanding of various UAV relaying schemes, in this section, we first outline some basic implementation issues of UAV-enabled relaying networks from a technical perspective. Then, we provide a comprehensive review of recent advances in aerial relaying schemes, which are categorized in their main contributions, i.e., novel relaying schemes, deployment of UAV relay(s), security and reliability and resources optimization of UAV relaying networks. Specifically, to show the developments in the area of aerial relay networks, the key factors and performance metrics of the related literatures are compared and discussed. Meanwhile, open challenges of each covered area are provided for a deeper treatment.

3.1. Technical details for aerial relay implementation

Some key implementation elements in UAV-enabled relaying networks are highlighted in Fig. 4.

Among various relaying methods, dual-hop relaying is a popular one by which the channel from a source node to a destination node is split into two, possibly shorter, links using a relay node. Multi-hop relaying can extend the network coverage and enhance throughput due to shorter hops, which can therefore guarantee network connectivity to locations that the general single-hop network cannot connect. Furthermore, more performance improvements in a multi-hop system can also be achieved via spatial diversity gains. Considering the use of a single UAV as a relay, the performance of the dual-hop relaying system can be improved through the trajectory/placement design of UAV, with a relatively low-complexity algorithm. On the other hand, multiple UAVs forming a single multi-hop link or multiple dual-hop links will bring more benefits, as well as challenges. In order to accomplish relaying tasks via UAVs cooperatively and efficiently and realize the communication of multi-platform and multi-task, the Internet of UAVs is employed as a future supplement to the UAV relaying communications. In such a scenario, the UAV group is networked for large-scale relaying communications, forming a multi-layer relay plane and providing flexible and reliable connections.

Despite advanced progress in relaying techniques, as well as the degree of spatial freedom brought by UAV relays, the implementation issues of UAV relay include the design of relaying schemes involving operation modes (half-duplex (HD), full-duplex (FD) or hybrid), relaying protocols (amplify-and-forward (AF), decode-and-forward (DF), compress-and-forward (CF), incremental relaying (IR), or buffer-aided), and access techniques (OMA or NOMA, as shown in Fig. 5), the relay selection according to various QoS requirements, the trajectory/placement design to take full advantage of the flexibility of UAV deployment, the security and reliability of A2G and A2A channels due to the high potential of free-space transmission and the joint resources optimization of power control, bandwidth allocation, time management and beamforming.

3.2. Novel UAV relaying schemes

Relaying is an extensively studied technique in terrestrial communication systems for throughput and reliability improvement, as well as coverage extension. Due to practical constraints such as limited mobility and wired backhauls, most relays in terrestrial systems are deployed in fixed locations, which we refer to as static relaying. However,

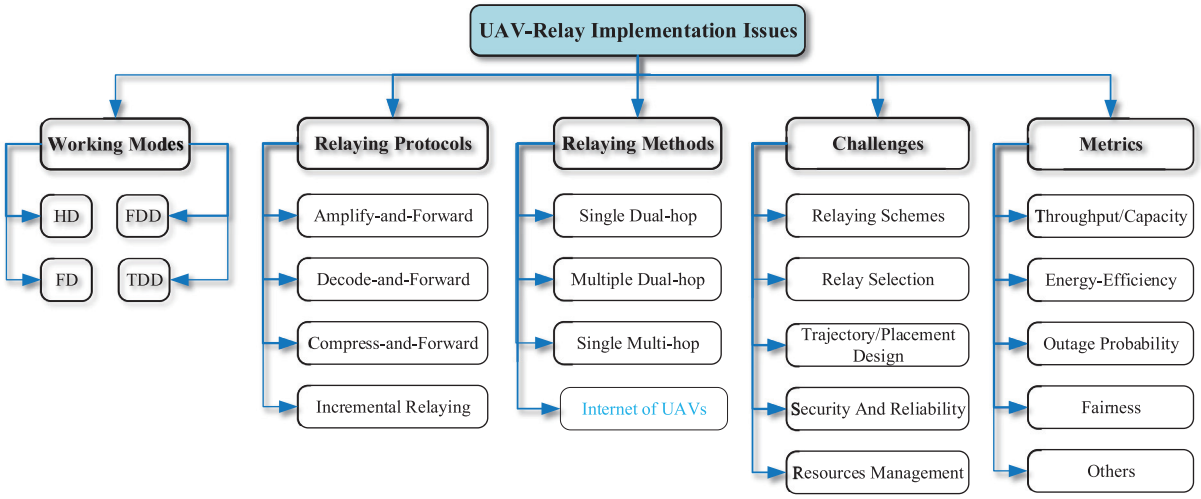


Fig. 4. Implementation details of UAV-enabled relaying networks.

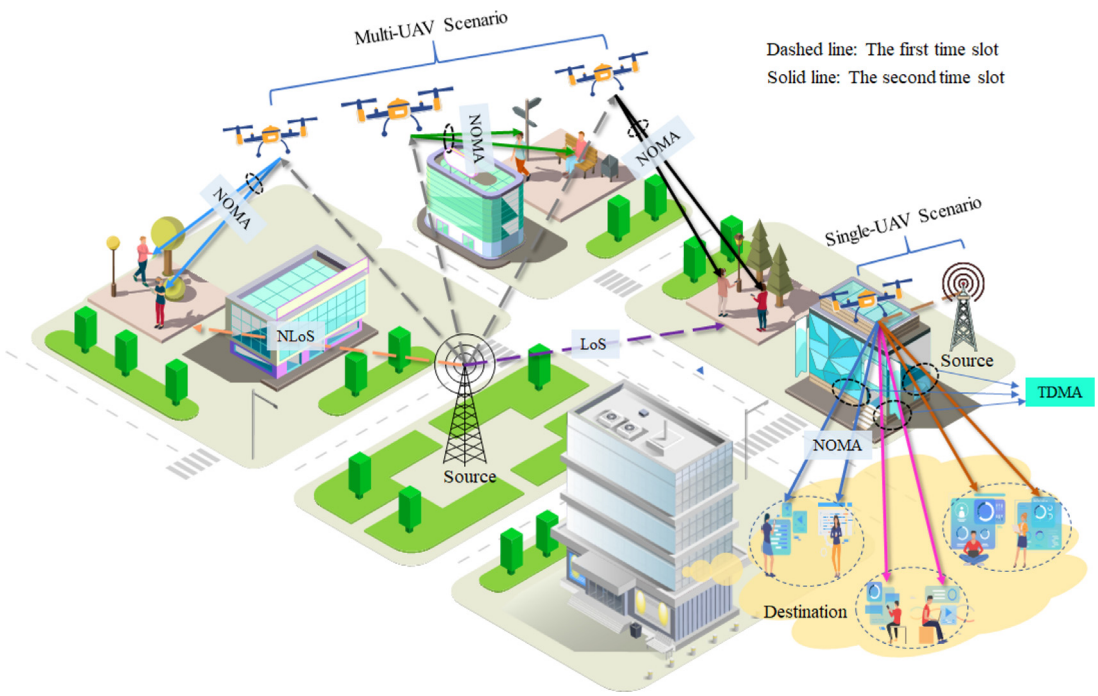


Fig. 5. UAV relaying networks with hybrid access protocols.

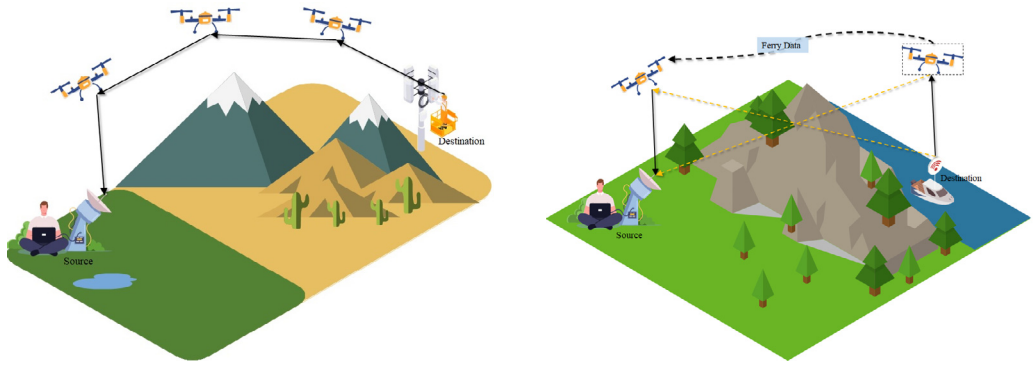
communications among GUEs are often subject to geographic location constraints. Through the prediction of UAV trajectories, it is possible to use the aerial relay to establish efficient LoS links. Particularly, multi-hop relay transmission technology can lay the foundation for multiple UAV relays.

3.2.1. State of the art

From the perspective of the number of UAV relays, the aerial relaying schemes are usually classified as dual-hop and multi-hop, as shown in Fig. 6. In [111], the end-to-end SNR was firstly maximized by evaluating the placement of UAVs, based on which two relaying setups of a single multi-hop link and multiple dual-hop links utilizing both AF and DF protocols were compared in terms of OP and BER. The effects of various important system parameters, i.e., UAV positions and source-to-destination distance, on the optimum settings were also examined in [111]. From the perspective of the operation modes of UAV relays, the aerial relaying schemes usually refer to AF, DF, CF or IR. Moreover, the buffer-aided relays in traditional relaying networks, by storing packets and transmitting them in favorable wireless conditions, have shown that buffering provides improved throughput, an increased stability region and better traffic load [112].

Despite traditional relaying schemes that combine multiple relaying and access techniques, the researchers have proposed novel relaying methods to further improve the performance of UAV relaying systems, as shown in Table 6. Moreover, considering the mobility and storage capabilities of UAVs, the authors in [113] implemented flying UAVs for data ferry, where the transmitter uploads data when UAVs approach, and then UAVs offload the data when approaching the receiver.

Specifically, UAVs that travel through cities and suburbs, such as airborne delivery systems, can be used for data relay and ferry for D2D communications, wireless backhauls, and cellular communications. Note that real-time information can be transmitted by aerial multi-hop relays, while the ferry model is usually used for non-real-time data transmission. For the design of dual-hop relaying, a store-then-amplify-and-forward (SAF) relaying protocol was proposed in [114],



(a) Multi-hop single link setting. (b) Dual-hop multi-link setting.

Fig. 6. Multi-UAV relaying systems.

in which the received signal from the source is stored in a buffer at the UAV, then the UAV amplifies and forwards it to the destination when flying closer to the destination. Assuming the UAV relay works in FDD mode [114], timeslots scheduling for each data packet received/transmitted by the UAV was optimized by the alternating optimization (AO) method. The FD UAV relay was implemented in [115] to help a pair of BSs jointly communicate with two distant GUEs via relaying NOMA signals, and the closed-form expressions of OPs were derived considering imperfect successive interference cancellation (SIC) at the Rx. Moreover, a NOMA-based selective incremental relaying protocol was proposed in [116], where an access point transmits a superposed signal based on the NOMA principle to a UAV. As for the multi-hop relaying, an adaptive direction control scheme for the UAV relay equipped with multiple directional antennas, was proposed in [117] to enhance the throughput performance. In order to accurately optimize a UAV's antenna radiation direction, [117] enables each UAV to rotate relatively to the position of its neighboring UAVs according to the straight or orthogonal relaying methods. The cluster-head-driven uplink model for multiple-antenna UAV relay was presented in [118], which designed a recursive relay selection strategy based on the likelihood of symbol detection in the UAV cluster-head.

3.2.2. Open challenge

Although multiple novel and efficient UAV relaying schemes have been developed, their main concerns are the limitation of communication resource and the distance between the transmitting and end nodes. One domain that has gained considerable attention is coverage, where UAVs are required to provide wireless connectivity between certain users, and communication quality is a basic mission requirement in coverage. Considering the energy sensitivity of UAVs, the system of minimizing the transmission energy while guaranteeing the throughput is another solution. However, the UAV relaying scheme for search and rescue scenarios lacks research attention. In disaster scenarios, such as earthquake rescue, water rescue and mountain rescue, UAVs are configured to search for and detect users and provide simple communication services for them. Communication quality is no longer the most important service indicator. Instead, it has become the most important UAV relaying mission requirement to search for and serve as many users as possible as quickly as possible.

3.3. Deployment of UAV relay

While the overall performance of the conventional network with a static relay node mostly depends on the distance between the source and relay nodes and that between the relay and destination nodes, in the UAV relaying system, these distances may change depending on the turning radius and flight altitude of the UAV due to its mobility. Therefore, the UAV can serve essentially as *compensation of pathloss* between the source and destination. On the other hand, thanks to the inherent mobility feature of UAVs, the trajectory/deployment design, as an interference evasion mechanism, has great potential for *interference avoidance/mitigation* in UAV-aided relaying networks.

3.3.1. State of the art

In [119], multiple UAVs were utilized as moving relays among ground stations with disconnected communication links in the event of disasters, where a variable-rate relaying approach was proposed to improve the OP and information rate. In [120], a UAV was implemented as a DF relay to assist command information transmission between a controller and a distant robot, where the decoding error probabilities (DEPs) at the UAV and destination were minimized through jointly optimizing the location of the UAV and the block length for the source-to-UAV and UAV-to-destination phases, under the latency constraint. Since most works use the conventional block coordinate descent (BCD) method for trajectory optimization, which is very likely to get stuck at the initial point, in [120], a small perturbation was introduced in each iteration. In [121], the UAV was deployed to enhance the communications between a satellite and terrestrial UEs whose locations were modeled as PPP, and the 2D location of the UAV was obtained when the minimal average distance between the UAV and UEs was achieved.

To get more insights into the system behaviors in terms of the system coding gain and diversity order, [122] derived an asymptotic expression for the high-SNR OP, which was then minimized through the design of UAV 3D placement. [123] considered multiple solar UAVs as relays to improve wireless communications among multiple ships with/without direct links, where the weighted-sum of communication cost (related to the probability of connectivity) and energy absorption of UAV was minimized, through using the online model predictive control (MPC) to optimize the UAV trajectory. In [124], the UAV relay was deployed in multi-cell networks to enhance the cell-edge communications, where the DF UAV helps to forward information from adjacent BSs to UEs distributed in the joint edge coverage of multiple cells.

In the scenario with different kinds of interference and different numbers of UAVs, [125] optimized the dual-hop UAV relay deployment for the improvement of system SNR. In the context of multi-hop relaying, the minimum required number of UAVs and their optimal positions were theoretically determined. The interference from nearby UEs on the BS-to-UAV link was considered in [126], whose objective is to minimize the power consumption by jointly optimizing the 2D placement of UAV and the power allocation for the NOMA signal. User fairness was also considered in [126], where the BS transmits NOMA-based signals to its nearby served users and the UAV relay. After decoding the remote users' messages from the NOMA signal, the UAV simultaneously transmits them to the remote users with less power allocated to the user characterized by a better channel condition. Assuming the prior knowledge of the channel model and the exact

Table 6
Comparison of literatures about novel UAV relaying schemes.

Case	Ref.	Relay mode	Operation	CSI	Relay scheme	Description	Metric
Single dual-hop	[114]	FDD	AF	Exact/Time-varying	SAF	The received signal from the source is first stored at the UAV, then amplified and forwarded to the destination when the UAV flies closer to the destination.	Throughput
	[115]	FD	DF	Statistical/Static	NOMA-assisted UL/DL design	A pair of BSs jointly communicate to two UEs via a UAV relaying NOMA signals, in the scenario of imperfect SIC.	OP, Throughput
	[116]	IR	DF	Statistical/Static	NOMA-based SIRP	With the existence of direct link, that the relay decision is determined by both the received SINRs of Rx and UAV.	OP
Single multi-hop	[117]	FD	DF	Exact/Static	Adaptive UAV direction control	Each UAV optimizes its antenna direction, according to straight or orthogonal relaying method and the positions of nearby UAVs.	Throughput
	[118]	HD	DF	Statistical/Static	Cluster-Head-Driven Best-Link	A recursive maximum minimum distance relay selection strategy for a multi-way multi-antenna multi-access UAV relaying network with broadcast-channel.	Delay, BER

Table 7
Comparison of literatures about the deployment of UAV relay.

Motivation	Ref.	Deployment	Metric	No. of UAV	Operation	Relay mode	CSI	Feature	Algorithm
Compensate for pathloss	[119]	2D placement	OP & Throughput	Single	HD	DF	Statistical/ Time-varying	The allocation of UL and DL channels are adjusted by UAV's locations.	Offline
	[120]	2D placement	DER	Single	–	DF	Instantaneous/Static	Optimization for the data sizes of UL and DL.	Offline
	[121]	2D placement	Throughput	Single	HD	DF	Statistical/Static	Imperfect CSI and random user pairing for NOMA signals.	Offline
	[122]	3D placement	OP	Single	–	DF	Statistical/Static	The size and center of coverage region are determined by users' locations.	Offline
	[123]	2D trajectory	Cost	Multi	–	–	Statistical/Static	Energy harvesting for UAVs; Optimize UAV trajectory online in a dynamic environment.	Online
	[124]	2D trajectory	Throughput	Single	FDD	DF	Statistical/ Time-varying	The enhancement of cell-edge communications.	Offline
Interference avoidance	[125]	2D placement	Capacity	Single & multi	–	DF	Exact/Static	Management of various interference from the environment; Dual- and multi-hop UAV relaying.	Offline
	[126]	2D placement	EC of UAV	Single	HD	DF	Instantaneous/Static	Interference from UEs not served by the UAV was considered; Channel with better CSI and less power.	Offline
	[127]	3D placement	Capacity	Single & multi	–	AF	Instantaneous/Static	Considering the external interference; The channel model and exact positions of GUEs are prior unknown.	Online
Tradeoff	[128]	2D placement	Cost	Single	–	Hybrid	Instantaneous/Static	The fine-grained degree of obstruction in AG propagation.	Offline

positions of UEs were unknown, [127] designed a model-free online learning algorithm to optimize the 3D placement of the UAV relay for single UAV and multi-UAV scenarios, respectively.

To strike the best trade-off between minimizing propagation distances and discovering good propagation conditions, the UAV in [128] operates as a flying wireless relay to provide coverage extension for a BS and deliver a capacity boost to a user shadowed by obstacles. Instead of static models of propagation links, [128] proposed a nested segmented propagation model for the UAV-UE link to capture the fine-grained degree of obstruction in A2G propagation. Based on the proposed channel model, a general cost function of the UAV-UE and BS-UAV channel gains was defined, and then minimized through the UAV 2D trajectory design. The characteristics of the above-mentioned UAV relaying works have been listed in Table 7.

3.3.2. Open challenge

In terms of UAV deployment, there is a need for new solutions for the optimal 3D placement of UAVs while accounting for A2G channel characteristics and UAVs' unique features. Besides, there is a need to study how UAVs must be deployed in coexistence with cellular

Table 8
Comparison of literatures about security and reliability for UAV relaying networks.

Metric	Eve		Ref.	Relay			Scenario	CSI		Secure QoS constraints
	No.	Location		No.	Relay mode	Operation		Legitimate	Wiretap	
SC	Single	Fixed	[129]	Single	FDD	DF	SISO with buffer-aided relay	Instantaneous/ Perfect	Instantaneous/ Perfect	Information-causality constraint
			[130]	Single	TDD	DF	SISO with tradeoff between secrecy rate and UAV's EC.	Statistical/ Perfect	Statistical/ Perfect	Information-causality constraint
			[131]	Single	FDD	–	SISO with a UAV jammer	Instantaneous/ Perfect	Instantaneous/ Perfect	Information-causality constraint
			[132]	Single	–	–	SIMO with local caching	Instantaneous/ Perfect	Instantaneous/ Perfect	Secrecy rate constraint
			[133]	Multi	–	–	SIMO with cache-enabled D2D communications	Instantaneous/ Perfect	Instantaneous/ Perfect	Transmission rate constraints
			[134]	Single	–	DF	SISO	Instantaneous/ Perfect	Instantaneous/ Perfect	Analysis
			[135]	Multi	HD	DF	SISO with EH at the source and relay	Statistical/ Imperfect	Statistical/ Imperfect	Analysis
		Dynamic	[136]	Multi	–	DF	SISO with a satellite as the source	Statistical/ Imperfect	Statistical/ Imperfect	Analysis
	Multi	Dynamic	[137]	Single	TDD	DF	SISO with the source performing SWIPT	Instantaneous/ Perfect	Instantaneous/ Imperfect	EH constraint
			[138]	Single	FD	AF	SISO with EH at UAV and the source as jammer	Statistical/ Perfect	Statistical/ Perfect	EH and EC constraints
SOP	Multi	Dynamic	[139]	Multi	HD	DF	SISO with jamming UAVs	Statistical/ Perfect	Statistical/ Perfect	Analysis
	Single	Fixed	[140]	Single	–	–	SISO with 3D beamforming design	Statistical/ Perfect	Statistical/ Perfect	N/O

networks while considering mutual interference between such aerial and terrestrial systems. In addition, the impact of inter-cell interference should be considered when simultaneously deploying multiple UAVs. Furthermore, trajectory optimization in UAV-enabled wireless networks requires capturing the coupling between mobility and various QoS metrics, i.e., the mobility patterns of ground users for maximizing the coverage performance, obstacle-aware trajectory optimization under delay and energy constraints. Finally, for cellular-connected UAV-UEs, optimizing the trajectory while avoiding interference to ground users and being aware of the beamforming of ground base station antennas is another open challenge.

3.4. Security and reliability of UAV relaying networks

Although the use of UAVs in wireless networks can bring many benefits, such as lower delay and higher rates, UAV-aided communications also face more serious security issues than traditional communication systems. This is because the transmitted data are more prone to be intercepted by malicious eavesdroppers due to the higher probability of a LoS connection. Research on physical layer security (PLS) transmission in conventional wireless networks has been widely studied to improve the secrecy capacity. However, in UAV-relaying networks, the high-mobility of UAVs can be utilized to establish better channel links with users and weaken the channel gain of an eavesdropping link by flying close to legitimate users or far from the eavesdropper. Additionally, since the LoS channel gain depends only on the link distance, the UAV can precisely obtain the channel gains of all eavesdroppers if their positions are known. This thus resolves the issue of CSI estimation of wiretap links in traditional networks.

3.4.1. State of the art

The state-of-the-art PLS designs in UAV relaying networks are mainly focused on trajectory design, resource allocation and cooperation, with the design objectives of secrecy capacity (SC) maximization and secrecy outage probability (SOP) minimization. The related works are listed in Table 8.

Illegal eavesdropping: In the UAV-assisted relaying system with one eavesdropper overhearing the relaying phase, the SC was investigated in [129–131,141], where the information-causality constraint is imposed to ensure that the relay cannot forward the signal that has not been decoded. Specifically, the authors in [141] optimized the power allocation for the Tx and UAV in each timeslot, then they extended [141] in [129] with UAV trajectory design involved. To maximally exploit the fully-controllable UAV mobility, the TDD operation is adopted by the HD UAV in [130], where the phases of receiving and forwarding information were dynamically scheduled following the UAV trajectory design, to strike a better tradeoff between the secrecy rate and the energy consumption of UAV. Two UAVs were introduced by [131], with one acting as the relay node and the other transmitting jammer signals to confuse the Eve. In addition, the transmit powers and trajectories of the aerial relay and jammer were jointly optimized to maximize the average SC. [132] studied the security performance for UAV-relaying systems with local caching, in which for each pair of users who have cached the file that is required by the other, the UAV can simultaneously broadcast the two files to them to disrupt the eavesdropping, while the secrecy rate of those users without caching is maximized when the secrecy rate of the caching users is satisfied.

Furthermore, at different timeslots, the UAV relay can ferry files from the BS or broadcast different kinds of files to UEs according to their requirements. In [133], caching capacity was enabled at UAV relays and D2D transmitters. [133] assumed that if the caching of UAVs

or user devices has the requested content locally, a UAV downlink or a local D2D communication link is respectively established; otherwise, UAVs fetch the desired contents from the BS and then deliver them to users. Considering the fairness among users, the minimum secrecy rate among users was maximized in [133], where UAVs can fly close to each legitimate user to improve the channel quality and simultaneously move away from the eavesdropper to relieve the information leakage; at the same time, the transmit powers of UAVs are adjusted to reduce the co-channel interference. Through theoretical analysis, the authors in [134] derived the closed-form approximation of the intercept probability and the ergodic secrecy rate considering the variations of LoS and NLoS probabilities and shadowing effects. [135] proposed a cooperative secrecy transmission mechanism, where the transmit signal of the primary user was used as energy to drive secondary user nodes and multi-relay nodes to send signals. Considering the unexpected noise brought by multi-relay nodes, a UAV relay selection scheme with secrecy capacity maximization criteria was proposed in [135]. The closed expressions of SOP and the probability of non-zero secrecy capacity were derived, followed by the analysis of asymptotic SOP in the high-SNR case. In [136], a secure 3D UAV relaying scheme was proposed for hybrid satellite-terrestrial networks, in the presence of an aerial eavesdropper lying around the serving UAV relay in a circular plane.

The authors in [137] investigated the secrecy performance of mmWave SWIPT UAV-enabled relaying networks, where multiple eavesdroppers follow an independent homogeneous Poisson point process (HPPP), and each of them adopts the maximal-ratio combining (MRC) protocol to process confidential information intercepted from both UL and DL. Specifically, in [137], the received signal of Rx was split into two power streams, with one for energy harvesting and the other for information processing. Using a 3D antenna-gain model, [137] derived the closed-form expression of the lower bound of the average SC that is used to jointly optimize the transmit powers of Tx and UAV, power splitting ratio and UAV's horizontal location. In [138], the UAV employs the power splitting (PS) scheme to simultaneously receive information and harvest energy from the Tx, while the Rx simultaneously receives information from the UAV and imposes interference on Eves.

Focusing on the SOP optimization, [139] considered a mmWave communications network with multiple UAVs working as relays and jammers, and assumed multiple eavesdroppers randomly distributed on the ground. Specifically, the locations of relaying UAVs, jamming UAVs and ground eavesdroppers were modeled as independent PPP in [139], contributing to the proposed opportunistic relay selection scheme. The authors in [140] considered a system consisting of a multi-antenna UAV-relay, a single-antenna BS and a multi-antenna destination, as well as a multi-antenna Eve with a fixed location. When both the destination and Eve employ MRC to receive signals from the UAV-relay and Tx, [140] also derived the expression of the secrecy OP, which is minimized by jointly optimizing the 3D beamforming and 2D trajectory of the UAV, and it has been concluded that the system with uniform circular arrays (UCAs) outperforms that with uniform linear arrays (ULAs).

The above works considered passive Eve(s); however, the active Eve that transmits jamming signals to legitimate users is unfortunately neglected. What is worse, most of them assumed the Eve's location was perfectly known, while such an assumption is not always valid since a passive eavesdropper would not reveal its location information to the legitimate nodes. Moreover, most of them focused on the security of DL (UAV-to-Rx), while the UL (Tx-to-UAV) is also vulnerable due to the broadcast nature of radio signals.

Legitimate eavesdropping: Recently, wireless information surveillance, as a new paradigm shift in wireless security, has received growing attention for its effectiveness in dealing with the suspicious communication initiated by some unkind people, such as terrorists and commercial criminals. For example, [142] employed multiple UAVs, and each hovered over one specific relay for proactive eavesdropping in the suspicious AF multi-relay system, where the suspicious Tx based on opportunistic relaying protocol selects one best suspicious relay to communicate with the suspicious destination. Meanwhile, in [143], a suspicious UAV relay helps one suspicious Tx communicate with one suspicious Rx in the presence of a ground legitimate monitor. Specifically, [142] described a system where the suspicious Tx selects one best suspicious relay from multiple energy-constrained suspicious relays to forward suspicious information to the suspicious Rx, and each UAV hovers over one terrestrial relay to implement possible jamming or eavesdropping. The numbers of jamming and eavesdropping UAVs were optimized to maximize the eavesdropping success probability, and the authors found a tradeoff that jamming with more UAVs helps decrease the suspicious rate and enables the remaining UAVs to overhear more clearly, but degrades the precondition of successful eavesdropping. The legitimate ground monitor in [143] aims to eavesdrop on the suspicious communication between a suspicious Tx and a suspicious Rx, which is established via a suspicious UAV-enabled DF relay. The transmitting and forwarding signals can be overheard by the monitor, and the monitor also imposes jamming noise on the relay and Rx. However, to maximize the communication rate, the UAV relay can dynamically adjust its location, which makes it difficult to figure out the exact jamming effects on air–ground links. To tackle these challenges and maximize the eavesdropping rate, the optimal UAV deployment and the sub-optimal jamming power were immediately derived by the designed heuristic scheme.

3.4.2. Open challenge

Many works have identified both the physical and cyber threats of UAV relaying systems and discussed the security properties required for their critical operational environment. These works propose solutions to the communication security of UAV relaying networks, while the control security of UAV relays has not been covered. What is more, adversaries are capable of launching advanced attacks such as side-channel attacks, fault injection attacks and software attacks to retrieve desired information from a drone. To deal with such attacks, drones should be equipped with superior cryptographic mechanisms and secure key management. In terms of data security, much effort has been put into drones' data collection and sharing, while data aggregation is another emerging challenge. Moreover, providing secure data aggregation that also achieves confidentiality and access control is another potential future research topic.

3.5. Resources optimization for UAV relaying networks

In this section, we will discuss the critical aspect of resource optimization for UAV relaying networks. As UAVs continue to play a significant role in various communication applications, including data collection and information broadcasting, it is essential to address the challenges associated with efficient resource management. The efficient allocation and utilization of resources, such as energy, bandwidth, and computation power, are crucial to ensure the reliable performance and extended operation of UAV-enabled networks. We will explore various techniques and strategies aimed at optimizing resources while maintaining network efficiency and meeting the specific requirements of different UAV communication scenarios.

3.5.1. State of the art

Various resources optimization schemes designed for UAV-relaying systems are classified according to their design objectives, as shown in Table 9.

OP: The analytical expression of the OP of UL (Tx-UAV) was derived in [144] where the distances of Tx-UAV and UAV-Rx, as well as the transmit powers of UAV and UE, were jointly optimized for the OP minimization. In [145,147], the NOMA signals are transmitted from the UAV to UEs, and the OP is related to both successful SIC probability and minimum SINR threshold. Specifically, the DF multi-antenna UAV

Table 9
Comparison of literatures about resources optimization for UAV relaying networks.

Objective	Ref.	UAV relay		Optimization items					Mathematical tool	Feature
		Operation	Relay mode	Trajectory/Placement	Power-control	Bandwidth allocation	Time management	Beam pattern		
OP	[144]	HD	AF	3D	✓				Optimization theory	Performance enhancement for UL.
	[145]	HD	DF		✓				Analysis	SIC capability and the no. of UAV's antennas.
	[146]	FDD	DF		✓	✓			Optimization theory	FDMA for ULs and NOMA for DLs; Successful SIC probability.
	[147]	FD	DF	3D			✓		Optimization theory	The feedback loop self interference of UAV.
	[148]	–	DF		✓		✓		Analysis	Applicable to diverse channel models.
TH	[149]	Hybrid	–		✓	✓			Optimization theory	Two-way relaying.
	[150]	FDD	AF	2D	✓				Optimization theory	
	[151]	–	DF	2D	✓				Optimization theory	Multi-hop relaying.
	[152]	–	AF	No. of tiers					Optimization theory	
	[153]	HD	AF			✓			Online learning	Relay assignment.
	[154]	HD	AF	3D		✓			Reinforcement learning, Online learning	
	[155]	–	DF	3D					Analysis	Mixed RF-FSO Channels.
	[156]	FD	DF	2D					Optimization theory	
	[157]	FD	DF		✓			✓	Optimization theory	Beamforming.
	[158]	FD	–	3D	✓			✓	Alternating interference suppression	
	[159]	FD	–	2D	✓				Optimization theory	Spectrum sharing.
	[160]	–	DF	1D	✓	✓			Optimization theory	
Fairness	[161]	TDD	AF	2D	✓		✓		Optimization theory	Multi-pair of Tx and Rx served by a UAV.
	[162]	AF	–	2D	✓				Optimization theory	ECs and the throughput respectively act as the constraint and objective.
	[163]	FDD	DF	2D	✓	✓			Optimization theory	Two-way relaying.
	[164]	HD	DF	2D	✓		✓		Optimization theory	Timeslots scheduling for DLs and ULs.
	[165]	–	–	2D		✓	✓		Optimization theory	Direct links involved.
EE	[166]	HD	DF	2D	✓				Optimization theory	Neglect UAV's EC for communication.
	[167]	HD	AF		✓			✓	Optimization theory	Beamforming at the satellite and UAV.
	[168]	–	AF	2D	✓				Optimization theory	Fuel-powered UAV.
	[169]	–	–	3D	✓				Optimization theory	Weight sum of EE of communication and WPT.
	[170]	–	AF	2D					Optimization theory	UAV flies close to S or D in different slots.

TH: throughput.

was employed in [145] to help the transmission between two pairs of Tx-Rx in a NOMA manner, and the closed-form expression of OP was derived, which is determined by the power allocation for the UL and DL, SIC capability at the UAV and UEs, and the number of antennas at the UAV. The OP in [146] is determined by the transmission rate of UL and DL and the SIC probability at UEs, which was minimized by jointly optimizing bandwidths (for FDMA uplinks and the NOMA downlink) and power allocations (for uplink and downlink signals). Considering the SINR threshold of the UL and DL and SWIPT at the UAV-relay, the system OP was minimized in [147,148]. Furthermore, the time duration for WPT and EH was optimized in [147], while the power allocation for information decoding and energy harvesting was optimized in [148].

Throughput: Compared with conventional static relaying, mobile relaying offers a new degree of freedom for performance enhancement via careful relay trajectory design. The basic model for a UAV-enabled mobile relaying system was presented in [171], and a throughput maximization problem was formulated to optimize the relay trajectory and the source/relay power allocations in a finite time horizon, subject to practical mobility, transmit power and information-causality constraints. Research results show that compared with conventional static relaying, a significant throughput gain is achieved by the proposed mobile relaying design, which shows the great potential of mobile relaying for future wireless communication systems.

[149] investigated a UAV-aided two-way relaying network, where a UAV-relay helps multiple pairs of users exchange their information within one timeslot, and each pair of users is allocated a slice of communication bandwidth to avoid inter-pair interference. Through optimizing the bandwidth allocation for each pair of UEs and the transmit power of UEs, the sum-rate of each UE was maximized in the cases where the UAV works in FD mode with server self-interference (SI) and HD mode without SI, respectively. Moreover, the time-fractions for information transmitting and forwarding were also optimized for time-fraction (TF)- and HD-based transmission, which proved to be superior to the FD-based relaying scheme. In [150], an AF UAV-relay was employed to help facilitate two-way communications between a ground BS and a set of distant UEs over orthogonal frequency bands, and the sum-rate of the two-way communication links was maximized for single-UE and multi-UE cases, respectively.

For a multi-hop UAV relaying system, the end-to-end throughput was maximized in [151], via UAVs' 2D trajectory and Tx/UAVs transmit powers optimization. A number of UAVs were utilized as multi-tier relays between a number of users and a BS in [152], where the wireless propagation channels between users and BS were modeled by the Rayleigh product channel model. Based on this, the lower bound of ergodic capacity was derived in closed form, which was further maximized by optimizing the number of tiers for a given total number of UAVs.

Considering the time-varying CSI, as well as the dynamic competition relationship and interference among the source–destination pairs caused by the dynamic positions of UAVs, [153] investigated the opportunistic UAV relaying network, in which the problem of joint optimization of relay assignment and channel allocation was formulated. The aim was to maximize the long-term average total transmission rate. Specifically, a single-relay can serve more than one source–destination pairs in a TDMA manner. While the source node selects only one UAV relay for relaying. Unlike [153], where direct links always exist between each pair of source–destination, the probabilistic direct link was involved in [154]. Here the capacity of the relay network was maximized by optimizing the relay scheduling, channel scheduling and UAVs' 3D trajectories.

UAV-enabled relaying networks with mixed RF-FSO channels were respectively investigated in [155,156] to analyze the system throughput. Specifically, a UAV relay with multiple RF links accessing mobile users and an FSO backhaul link access to the BS was considered in [155], where the probabilistic LoS RF channels and the quality of FSO channel respectively depend on UAV's position and instability. Based on which, the ergodic sum-rate was respectively analyzed for the system with a buffer-aided UAV-relay and a non-buffer-aided UAV-relay. The impacts of UAV position, UEs' density, and weather conditions on the ergodic sum-rate were respectively illustrated. In [156], a UAV relay with a limited buffer utilizes a FSO link to receive information from a backhaul terminal, and uses a RF link to forward information to a user terminal and the UAV's 2D trajectory, velocity and acceleration were optimized to maximize the data rate of the UAV-user link.

Dual-hop relaying techniques have recently gained great interest in the context of cooperative wireless networks due to their numerous benefits over single-hop relaying, such as increasing signal reliability and extending the radio coverage. By inserting beamforming techniques into dual-hop transmission schemes, such advantages become even more pronounced, which makes them very promising for future wireless networks. In [157], a DF UAV-relay with multiple antennas flies a circular trajectory to help relay information for a pair of source–destination, and the transmit/receive beamforming of the UAV and power allocation for the source and UAV were jointly optimized to maximize the instantaneous achievable rate of the destination. In [158], a FD UAV-relay was employed to compensate for the high pathloss and to mitigate SI at the UAV. A mmWave network with a pair of source–destination equipped with multi-antenna was investigated in [158], where the UAV position, analog beamforming for Tx transmitting and for UAV transmitting/receiving and power allocation for UAV and Tx were jointly optimized to maximize the achievable rate from Tx-to-Rx.

A spectrum-sharing mode for aerial relaying and terrestrial D2D communications was introduced in [159], where the FD UAV-relay equipped with two antennas simultaneously receives and transmits signals over the same spectrum, which is reused by D2D pairs. UAV's 2D trajectory and power allocation for the UAV, Tx and D2D UEs were jointly optimized to maximize the achievable rate of Rx and to enable the coexistence of UAV and D2D communications. The authors in [160] incorporated the NOMA protocol with the DF relaying protocol, to enable the DF UAV to enhance communications between the BS and cell-edge UEs. Specifically, the BS transmits data to the UAV in the first slot, and then the BS and UAV simultaneously transmit NOMA signals to the users via FDMA in the second slot. With the SIC technique and LoS and NLoS assumed for air–ground and BS-UE channels, cell-edge UEs receive and decode the dual information in the second slot. Besides, the sum-achievable rate of UEs was maximized by jointly optimizing the UAV height, orthogonal channel scheduling, and power allocation for Tx and UAV.

Fairness: The fairness issue is not negligible in communication networks since the air–ground channel changes when the UAV moves, and the UAV-relaying system may fail to meet the quality-of-service (QoS) requirements when the UAV flies to some positions. Under the constraints of transmit powers of the source node and UAV and UAV's velocity, the minimal rate among all possible DLs was maximized in [161], where an amplify-and-forward (AF) UAV-relay was employed to relay data for multiple pairs of GUEs in a TDMA manner, and the timeslots scheduling for communication pairs, power allocation for Txs and UAV and UAV's 2D trajectory were jointly optimized. To improve fairness and energy-efficiency, the minimum achievable instantaneous rate of Rx was maximized in [162], under the constraints of the transmission power budget and the UAV's flying speed, and then the sum-power consumption of the UAV and Tx was minimized for a given required data-rate, by joint transmit power and 2D trajectory optimization over a time horizon. Despite the constraints of transmit powers and the velocity and/or acceleration of the UAV, in data relaying, the information causality caused by backhaul links was studied in [163–165]. Specifically, a two-way relaying strategy was proposed in [163], where a UAV-relay helps bidirectional communications for a pair of GUEs; besides, the power allocation for the UAV and GUEs, UAV's 2D trajectory and bandwidth allocation for ULs and DLs were jointly optimized to maximize the minimum average rate among DLs. In [164],

a UAV-relay helps some ground BSs send information to remote GUEs in a TDMA manner, and the minimum average rate among DLs was maximized by jointly optimizing the UAV's 2D trajectory, timeslots scheduling, and power allocation for Txs and UAV in each timeslot. A DF UAV-relay in [165] was employed to enhance the communications from a BS and multiple GUEs in the urban area, with communication channels represented by the segmented ray-tracing model and the existence of direct links. [165] maximized the minimal rate among all UEs by jointly optimizing UAV's 2D trajectory, user scheduling, and bandwidth allocation for channels of UAV-UE, BS-UE, and BS-UAV in each timeslot.

EE: Due to the maximization of throughput resulting in the cost of UAV's energy consumption, the energy-efficiency defined as the trade-off between the throughput and the UAV's energy consumption for propulsion and communication has become an increasingly important indicator in designing UAV-enabled wireless communication systems. How to increase EE while meeting the requirements of wireless communication systems has drawn the attention of many researchers. In [166], a half-duplex DF UAV relay was implemented to help the communications from a BS to multiple GUEs, in FDMA manner, and the EE defined as the ratio of sum data rate of DLs to the power consumption of UAV was maximized by optimizing UAV's 2D trajectory, velocity, acceleration and power allocation. In [167], a multi-antenna UAV was employed to assist the communications from a satellite to a terrestrial node, and the angular information-based channel model was used to help design the beamforming for the satellite and UAV. By optimizing the beamforming design and transmit power of UAV, the EE defined as the ratio of the ergodic data rate of DL to the power consumption of UAV was maximized under the total power constraint and per-antenna power constraint at the UAV. Furthermore, by assuming that the satellite-UAV link undergoes correlated shadowed-Rician fading while the UAV-terminal link experiences correlated Rician fading, the closed-form expression for the output SNR was derived in [167]. [168] applied the probability LoS model proposed by ITU, which leads to the altitude of UAV constrained by the LoS air–ground channel, and then the system EE defined as the ratio of the average data rate of DL to the average EC of UAV, was maximized by jointly optimizing the power allocation and UAV's 3D trajectory. A UAV relay performs SWIPT in [169], where the weighted-sum EE during information transmission and wireless power transmission was maximized by optimizing the power allocation for Tx/UAV/power supplier and UAV's 2D trajectory. In [169], the EE for information transmission was defined as the ratio of data rate to the sum-EC of UAV and Tx, and the EE for WPT was defined as the ratio of harvested energy to the EC of the power supplier. Instead of the trajectory optimization for UAV, the initial and final locations of UAV relay were optimized in [170], to maximize the EE defined as the ratio of the data-rate of DL to the EC of UAV. Specifically, the UAV receives signals at the initial location, caches the received signal from Tx during a fixed duration and forwards it to Rx when the UAV flies closer to the Rx.

3.5.2. Open challenge

Resource scheduling aims to optimizing resource configurations based on meeting the fundamental application requirements. There are a great deal of resources in UAV-aided wireless networks, including sensing, communication, computation and actuation resources. Furthermore, actuation resource scheduling involves the control of the physical world, which makes optimizing control and resource scheduling problems more complicated. For example, as a typical actuation resource, mobility is indispensable for UAV networks. However, the mission-oriented mobile resource scheduling may face challenges in an uncertain environment. Moreover, efficient cooperation coordination, flight formation, and cluster formation are critical for maximum throughput and energy-efficiency when multi-UAV relay systems are deployed. Finally, adopting suitable frequency bands (e.g., WiFi, LTE bands) for UAV operations is an important design problem.

4. UAV as MEC server

Recently, UAV-enabled MEC architectures that include UAVs into MEC networks have been developed. In those designs, a UAV can be regarded as a user who needs to execute computing tasks, as a relay to help users offload computing tasks, or as a MEC server for performing computing tasks. Even in the wilderness, the desert, and complicated landscapes, where terrestrial MEC networks might not be efficient or seriously constructed, UAVs can be deployed with flexibility to support terrestrial MEC networks. Furthermore, the computation performance can be improved since there is a high possibility that short-distance LoS links exist for offloading computation tasks and downloading computation results. To be specific, the UEs have significantly benefited from MEC by offloading their computing- and latency-intensive tasks to the MEC server (MECS). However, before the UE makes its offloading decision, factors like the battery and task queuing state at the UE, the computing capacities of the UE and UAV, the offloading channel condition, and the QoS requirement should be carefully evaluated.

Understanding the recent scientific developments and technological problems of UAV-enabled MEC networks is vital for realizing the advantages and facilitating the wide range of practical applications of UAV-enabled MEC networks. For this purpose, this section first introduces some technical background before presenting a comprehensive overview of the most recent research activities in the field of UAV-enabled MEC networks. We categorize UAV-empowered MEC networks based on three scenarios where single-UAV, multi-UAV and cooperative UAVs are employed as aerial MEC servers. To forth the development of UAV networks, UAVs benefiting from edge computing are also included in this section. The open problem existing in each application scenario is also summarized to shed light on future opportunities, as done in this section.

4.1. Technical details for aerial MEC implementation

Some key implementation elements in UAV-enabled MEC networks are highlighted in Fig. 7

For the partial offloading mode, the UAV may dynamically and flexibly assign its communication resources for offloading computation tasks and its computing capacities for performing local execution, based on the CSI. Since task offloading and local computation are carried out independently by the UAV under the binary computation mode, the flexibility of the resource allocation method will be constrained. In this regard, it is possible that the partial offloading mode will result in better computing performance than the binary computation mode. However, in the partial offloading mode, a more complicated circuit and protocol is needed. Additionally, both the design of UAVs and the specification of computing tasks can influence the choice of operation modes.

In UAV-enabled MEC networks, resources allocation is crucial because of UAV battery concerns and trajectory restrictions. In addition to enhancing computing performance, an effective resource allocation scheme can help ensure the UAV-empowered MEC systems operate reliably and economically. Additionally, a reasonable trajectory can be developed to strike a reasonable balance between the computing performance and the UAV operational costs. The allocation of resources in the UAV-enabled MEC networks can be managed to accomplish a variety of goals, including the maximization of computation bits, the minimization of energy consumption, the improvement of computation efficiency, the decrease in costs and completion time, and the consideration of fairness.

Without loss of generality, for the computation offloading and resource allocation, the formulated optimization problem is non-convex and difficult to solve; thus, researchers usually convert it into a more tractable form through a series of mathematical manipulations. Then, several sub-problems are decomposed from the original problem and solved via optimization methods. Finally, an iterative algorithm is

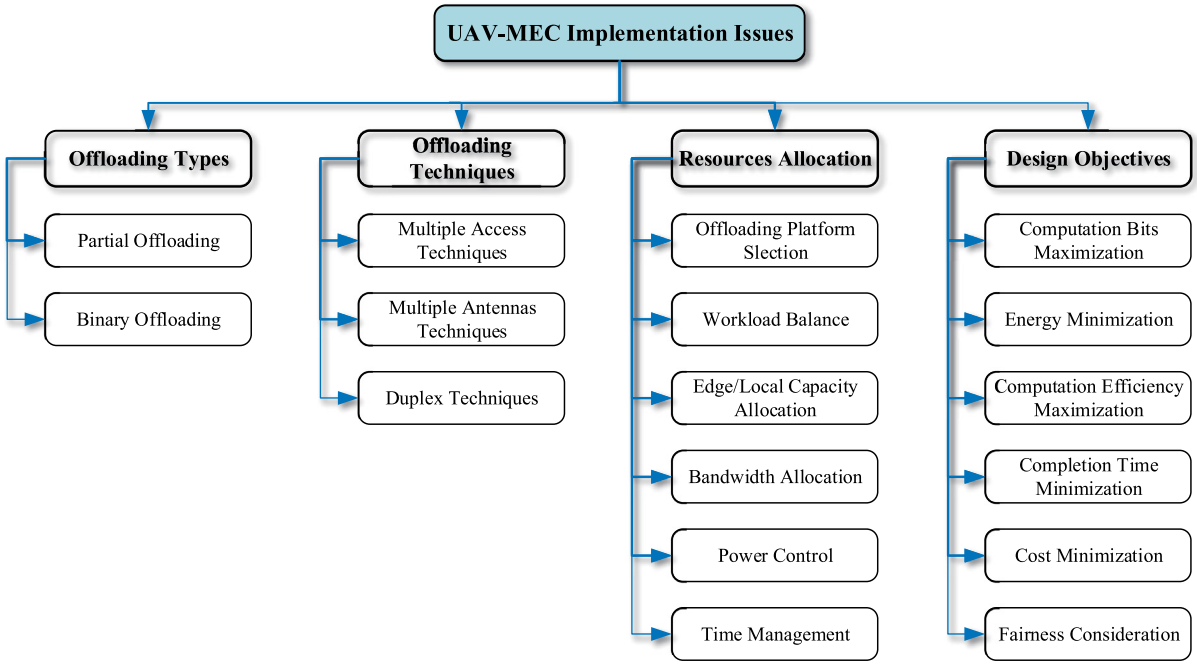


Fig. 7. Implementation details of UAV-enabled MEC networks.

developed to obtain a near-/sub-optimal solution, followed by an analysis of the algorithm complexity. Fig. 8 illustrates the general process of solving the formulated problem for UAV-enabled computation offloading systems.

4.2. Computation offloading to single-UAV

In this section, we will delve into the concept of computation offloading to a single-UAV setup. As the demand for real-time processing and low-latency communication increases in various UAV applications, offloading computation tasks to a UAV can provide an effective solution for resource-constrained devices on the ground. By utilizing the computational capabilities of a single UAV, ground devices can efficiently allocate their resources and focus on other critical tasks. We will discuss the potential benefits, challenges, and strategies associated with computation offloading to a single-UAV system, as well as explore how it can improve overall network performance and enhance the user experience.

4.2.1. State of the art

Specifically, in Table 10, we compare the literatures investigating the single-UAV enabled edge computing system, and illustrate the optimization items in Table 11.

With the objective of minimizing the sum-EC of all UEs while ensuring the successful task computation during a time block, [172] jointly optimized UAV's 2D position, timeslots allocation and the portion of offloaded tasks for each UE. The globally optimal solution was found by 2D search in the designed alternative optimization algorithm. In [173], the ECs of UEs and UAV were respectively considered as the objective function and constraint. The formulated mixed-integer nonconvex problem was solved by iteratively solving three subproblems corresponding to the task offloading decision, transmission bit allocation during each timeslot, and UAV's trajectory. Guo et al. in [174] investigated a UAV-enhanced edge scenario, where the UAV flies horizontally above those places with a high density of terrestrial UEs. The task offloading decision-making problem under the binary offloading paradigm was studied and solved via block coordinate descent (BCD) and SCA methods, aim to minimize the overall EC of UEs under the constraints of the maximum permissible task latency and the energy budget of the UAV. To convert the formulated mixed-integer nonconvex problem into a better tractable form, Hu et al. in [175] introduced auxiliary variables, and the resulting problem was solved with the proposed penalty dual decomposition-based algorithm, which contains two loops. To reduce the algorithm complexity, a simplified l_0-norm algorithm was further developed to minimize the sum EC of UEs, which was also extended to an average delay minimization problem. The TDMA-based workflow structure for the UAV-enabled MEC network was studied in [176], which allowed parallel transmissions and executions for different devices during the same timeslot. Specifically, the UAV hovers at a fixed location for a time duration, during which one IoT device could upload computation bits, and the service flow for a device contains three stages, i.e., UAV powering devices, data uploading and task computing. Thus, the weighted sum of UAV's energy consumption for computing, powering and flying was minimized under the energy constraints of IoT devices, latency requirements and UAV capacity.

In [177], the UAV has to compute and cache tasks offloaded from GUEs. At the same time, to minimize the EC of UAV for flying, computing and caching, the portions of a task offloaded to and cached by the UAV, as well as the UAV's 3D trajectory and flight time duration were jointly optimized. Specifically, the energy models and flight constraints of fixed-wing and rotary-wing UAVs were both analyzed in 3D spaces. Zhan et al. in [178] formulated two optimization problems for minimizing the EC and mission completion time of the UAV, which were respectively reformulated to be a tractable form that was solved via the aid of path discretization approximation technique and SCA-based algorithms. Subsequently, the Pareto-optimal solution was studied to balance the tradeoff between UAV's EC and mission completion time. The objective of [179] is to maximize the sum of bits offloaded from all UEs to the UAV, through jointly optimizing user association, the UAV's trajectory and the uploading power of each user, and they correspond to three subproblems that were iteratively solved with integer programming and successive convex optimization methods. The authors in [180] aimed to maximize the weighted-sum of offloading rates under the EH constraint of UEs, and both the partial and binary offloading models were investigated. By using the BCD method, the closed-form expressions for the optimal CPU frequencies, offloading times, and transmit powers of UEs were respectively derived. Moreover, the optimal selection scheme for whether UEs choose to locally

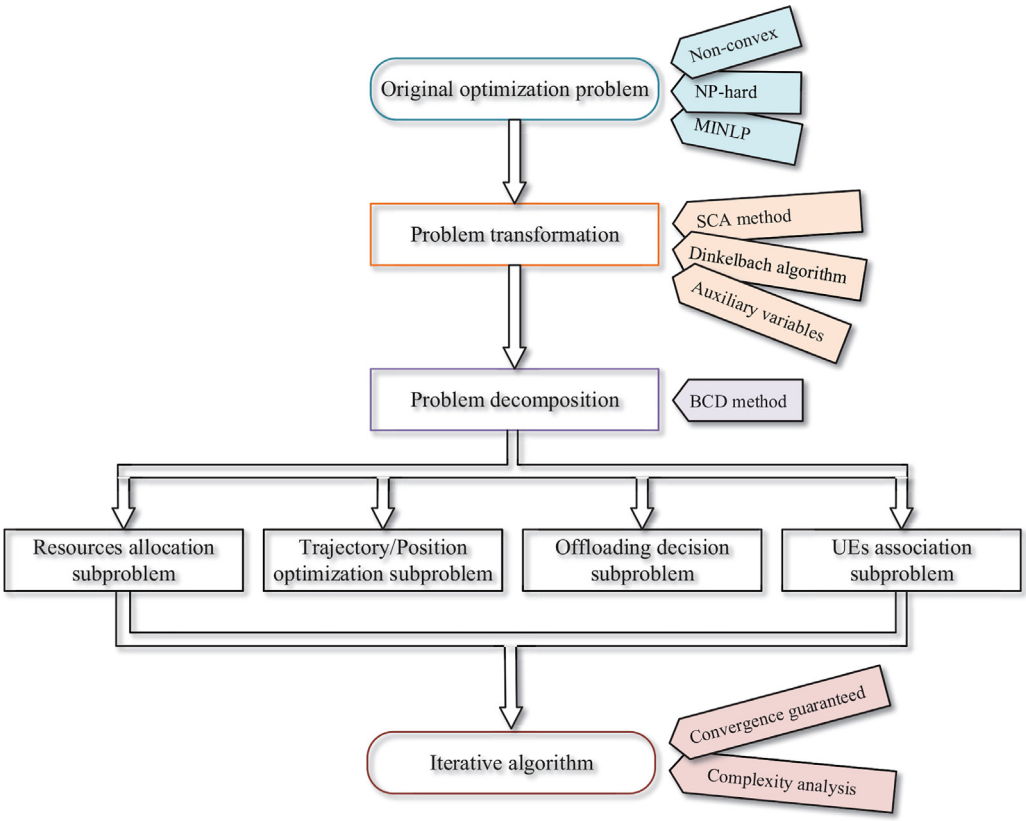


Fig. 8. The process for solving the computation offloading problem with optimization theory.

compute or offload tasks was proposed for the binary computation offloading mode.

The research results in [181] have shown that there exists a tradeoff between system utility and task queue stability. Specifically, the task queue at the UE was analyzed with the Lyapunov-based approach, then the computation tasks to be processed by each user were established as a stochastic queue model. Then, the weighted sum of UEs' and UAV's ECs was formulated as a non-convex problem, which was decomposed into three manageable subproblems, corresponding to stochastic computation offloading, CPU-cycle frequency allocation, and UAV 2D trajectory optimization. A combination of the alternating direction method of multipliers (ADMM) algorithm, the interior point method and CVX solver was employed to iteratively solve the subproblems. The research results in [182] show that the NOMA channel model performs better than the OMA model in terms of energy-efficiency, which was defined by the ratio between the overall offloaded computation bits and the EE of the UAV. Since the formulated EE maximization problem is a non-convex fractional programming problem, the successive convex approximation (SCA) technique and Dinkelbach algorithm were exploited to transform it into a solvable form, which was further decomposed into sub-problems corresponding to the power allocation for UEs and the computation load allocation and UAV trajectory planning for the UAV.

To ensure Min–Max fairness among multiple UEs or to obtain a narrow gap between the maximum and minimal EC of UEs, [183] formulated a more tractable non-convex problem by introducing auxiliary variables. This problem was decomposed into two convex subproblems, and solved by the proposed iterative joint task data allocation and trajectory optimization algorithm. The fairness issue in terms of EC was also investigated by the authors in [184], where the UAV's 2D trajectory, the volume of offloaded bits, and the computing resource allocated to each UE were jointly optimized. They first introduced auxiliary variables and utilized the structural characteristics of the formulated problem to convert the original problem into convex sub-problems that were iteratively solved. Moreover, to improve the algorithm complexity, the UAV was assumed to be fixed at a point, and then the fixed location was optimized.

4.2.2. Open challenge

In contrast to the management of traditional terrestrial MEC networks, resource optimization in UAV-empowered MEC networks involves computing and communication processes, as well as the flying processes of UAVs. To design, analyze and optimize UAV-enabled MEC networks, various mathematical tools are adopted, in which the optimization theory has been widely used. The researchers can usually choose one or more favorable methods from various convex optimization methods. However, user association and resources scheduling optimization usually lead to challenging mixed-integer programming problems that require advanced mathematical techniques.

4.3. Computation offloading to multi-UAV

In this section, we delve into the concept of computation offloading to multi-UAV systems. As the demand for real-time data processing and analysis increases in various applications, leveraging the computational capabilities of multiple UAVs can significantly enhance overall system performance. Computation offloading distributes complex tasks among multiple UAVs, mitigating the constraints imposed by limited on-board resources and ensuring efficient execution of time-sensitive processes. We will examine the challenges, strategies, and potential benefits of employing multi-UAV systems for computation offloading, as well as the role of communication and coordination among UAVs to achieve optimal resource utilization and task completion.

4.3.1. State of the art

The research results in [185] have shown that the energy-efficiency performance of multi-UAV-enabled MEC system is better than that of its single-UAV counterpart, and the system performance can be further

Table 10
Comparison of literatures about single-UAV-enabled edge computing.

Ref.	Offloading type	Objective	Proposed solution	Conclusions
[172]	Partial	Min UEs' EC	2D search alternating optimization algorithm employing augmented Lagrangian active set method	• Alternating optimization algorithm matches to 2D search algorithm.
[173]	Binary	Min UEs' EC	Alternative algorithm employing BCD method	• UEs offload more bits as UAV flies closer; • Lower EC obtained with larger time period.
[174]	Binary	Min UEs' EC	Alternative algorithm employing BCD and SCA methods	• Energy saving can be achieved by smaller timeslot; • UEs offload more bits as UAV flier closer.
[175]	Partial	Min the sum of UEs' maximum latency during a period	Penalty dual decomposition-based algorithm and simplified l_0-norm algorithm employing auxiliary variables	• The Penalty dual decomposition-based algorithm characterized by lower latency but higher complexity; • The system latency obtained by minimizing the sum of maximum delay among all the users in each timeslot is smaller than the one obtained by minimizing the maximum delay among all the users in one period.
[176]	Full	Min UAV's EC	Iterative algorithm employing flow-shop scheduling methods	• UAV hovering time can be saved by multi-workflow structure; • UAV hovering time rises as power conversion efficiency decreases.
[177]	Partial	Min UAV's EC	Iterative algorithm employing BCD and SCA methods	• UAV flies closer to UEs in a smoother way; • Fixed-wing UAV adopts a longer trajectory and consumes more propulsion energy compared with rotary-wing UAV.
[178]	Partial	Min UAV's EC Min UAV's latency	Alternative algorithm employing path discretization and SCA techniques	• UAV's propulsion EC model has impact on system design; • The tradeoff between completion time and EC of the UAV.
[179]	Full	Max offloaded bits	Iterative algorithm employing integer programming and SCA methods	• Increasing maximum velocity increases offloaded bits and flight energy.
[180]	Partial Binary	Max weighted-sum of UEs' computation bits	Two alternative algorithms for partial and binary offloading models, respectively	• The no. of timeslots has larger effect on algorithm complexity than the no. of UEs; • Offloading decision depends on the tradeoff between achievable computation rate and operation cost;
[181]	Full	Min weighted-sum of UEs' and UAV's EC	Iterative online algorithm combining alternating direction method of multipliers, interior point method and CVX solver	• There exists a tradeoff between system utility and queue stability; • The change of the total EC versus no. of timeslots is not linear.
[182]	Partial	Max UAV's EE	Alternative algorithm employing Dinkelbach algorithm, SCA and alternating direction method of multipliers	• The mechanical EC for optimizing trajectory in NOMA case is smaller than that in OMA case; • The EE was significantly improved compared to the scheme with circular trajectory.
[183]	Full	Min–Max UEs' EC	Iterative algorithm employing auxiliary variables	• UAV moves slow down when the distances between UAV and UEs are similar; • The trajectory of UAV is symmetrical.
[184]	Full	Min–Max UEs' EC	Two iterative algorithms employing auxiliary variables	• The largest EC of UEs becomes smaller as UAV's altitude becomes larger; • Trajectory optimization enjoys higher complexity and better energy performance than position optimization.

Table 11
Optimization items for single-UAV-enabled MEC system.

Variables	Ref.												
	[172]	[173]	[174]	[175]	[176]	[177]	[178]	[179]	[180]	[181]	[182]	[183]	[184]
Workload balance	✓					✓							
Offloading decision		✓	✓						✓				
Time allocation	✓				✓				✓				
Offloading rate		✓	✓	✓	✓					✓	✓	✓	✓
User association				✓	✓		✓	✓					
Power control								✓	✓		✓		
Local capacity allocation					✓		✓		✓	✓			✓
UAV flight period					✓	✓	✓						
User scheduling													
UAV trajectory		2D	2D	2D		3D	2D	2D	2D	2D	2D	2D	2D
UAV position	2D	2D											
Others					Tasks sequence	UAV caching							

Table 12
Comparison of literatures about multi-UAV-enabled edge computing.

Ref.	Features	Objective	Proposed solution	Conclusions
[185]	• A task is divided into sub-tasks with equal size for each timeslot;	Min weighted-sum of UEs' and UAVs' maximul ECs	Iterative algorithm leveraging greedy rules	• The offloading strategy optimization is more energy-efficient than the trajectory optimization; • UEs' ECs are sensitive to the difference between the no. of UEs and UAVs' computing resource;
[186]	• UAVs are dispatched to dynamic hot-spot areas identified by task intensity; • Theoretical analysis is given to verify the performance of proposed algorithm.	Max the no. of served tasks	A greedy online dispatching algorithm	• Dynamic tasks and nonuniform distribution of UEs result in unbalanced workloads among fixed BSs; • UAVs are dispatched closer to UEs;
[187]	• NOMA for UEs access to the same UAV, FDMA for UEs access to multi-UAV; • The probability of LoS/NLoS path depends on the evaluation angles of UAVs.	Min weighted-sum of UEs' and UAVs' ECs	Iterative algorithm leveraging SCA and quadratic approximation	• UE tends to offload tasks in the timeslot with better channel conditions; • The optimal number of UAVs depends on the network scale and the topology of UEs;
[188]	• The constraint of safe distances between UAVs and obstacles.	Max UEs' computation efficiency	Iterative algorithm leveraging the Dinkelbach method	• UAVs keep stationary above SMDs for several timeslots to execute the offloading tasks; • UAVs in MEC network with obstacles have a greater probability to fly at the maximal velocity.
[189]	• Multi-antenna UAV; • UAVs with overlapped coverage are allocated with orthogonal frequency;	Min weighted-sum of UEs' and UAVs' ECs	Iterative algorithm leveraging fuzzy c-means clustering	• Improvement of energy performance is limited by UAVs with high maximal computation capacity; • The energy performance of proposed algorithm performs better for low latency deadline, low maximal capacity of UAVs and high task complexity.
[190]	• Multi-UAV placement for workload balance; • Task scheduling in each UAV for improving task execution time;	Min UEs' average latency	Reinforcement learning-based task scheduling and differential evolution-based UAVs placement	• Benefit from task scheduling, the recent arriving small tasks can be processed immediately; • The tradeoff exists between latency improvement and communication cost.

improved by involving more UAVs, especially when the communication resources of UEs are limited. For the multi-UAV-enabled edge computing offloading system, we compare the related literatures in Table 12, in terms of their design objectives, features, solutions and conclusions.

Specifically, to ensure the EC fairness of UEs while considering the EC fairness of multiple UAVs, the weighted sum of the maximal ECs among all UEs and UAVs was minimized in [185], through alternately optimizing the offloading strategies and UAVs' trajectories. In [186], the 3D placements and communication ranges of UAVs were jointly optimized to maximize the number of tasks served by multiple capacity-limited UAVs, under a given latency deadline. Considering the mobility of UEs and UAVs, as well as the uniform distribution of GUEs, a greedy dispatching algorithm was proposed to determine the hover locations of UAVs in an online fashion. [186] also concluded that either mobile dispatching alone or the hybrid dispatching scheme is superior to fixed deployment.

With the objective of minimizing the sum of ECs of UEs and UAVs, Zhang et al. in [187] designed the radio and computation resources allocation and UAV 2D trajectory scheduling for the multi-user and multi-UAV system, in which the UEs share the same spectrum when offloading bits to the same UAV in a NOMA manner, while FDMA is adopted by different UAVs for interference migration. Besides, to perform SIC, the UAV ranks the associated MDs in ascending order of channel gain.

The computation efficiency, defined as the ratio of total computation bits to the sum-EC of UEs, was maximized by Zhang et al. in [188], through jointly optimizing the local CPU-cycle frequency, transmit power and spectrum resources, as well as 2D trajectories of UAVs. In [188], one user can only be associated with one UAV at one timeslot, which makes the formulated problem a MINLP problem; thus, the Dinkelbach method was adopted to eliminate the non-linear fractional problem and make it more tractable.

The UAV in [189] is equipped with a directional antenna, whose azimuth angle was optimized to achieve an adjustable beamwidth, and UAVs with overlapped coverage areas were allocated with orthogonal spectrums. Under the constraints of UAV's power, capacity and coverage, [189] proposed a fuzzy c-means clustering-based algorithm, and then an iterative algorithm was developed to solve the formulated non-convex problem with the goal of minimizing the weighted-sum of ECs of UAVs and UEs. Unfortunately, the interference among UEs when offloading computation bits and their accessing methods were not mentioned.

Considering the non-uniform distribution of GUEs, and with the goal of reducing the average latency for offloaded tasks, Yang et al. in [190] studied UAVs' 2D deployments, task scheduling, and the load balance for UAVs. Accordingly, they proposed a differential evolution (DE)-based multi-UAV deployment mechanism was proposed to reduce the transmission delay and balance the load of multi-UAV. They also developed and a deep reinforcement learning (DRL) algorithm for task scheduling in a certain UAV to reduce the queuing time of tasks and improve processing efficiency.

4.3.2. Open challenge

A multi-UAV network, UAV deployment and resource allocation are tightly coupled, and cooperative behaviors are of great significance. Besides, the UAV's connectivity is an issue that affects system performance. Most UAV-enabled MEC systems consider remote services for wildlife and forest monitoring and disaster response, while the interference has not been well studied. Furthermore, the battery life can affect the loss of connectivity. Thus, it is worthwhile to determine whether a UAV has any limiting factors that might hamper network connectivity. Methods based on artificial intelligence may resolve this problem since they allow the prediction of the network scenario in advance, which

Fig. 9. Various cooperation modes in UAV-enabled MEC systems.

can mitigate the risk of losing connectivity. Furthermore, with nodes having standard networking quality, topology can be established by incorporating networking strategies with path-planning techniques and dynamic UAV clustering-based offloading techniques [191].

4.4. Cooperative computation offloading

In this section, we will delve into the concept of cooperative computation offloading, a promising technique that enables enhanced performance and resource utilization in complex communication networks involving UAVs, edge devices, and other network nodes. With the rapid proliferation of data-intensive applications and increasing demand for real-time processing, traditional centralized computation models face significant challenges in meeting the stringent latency and energy requirements. Cooperative computation offloading addresses these challenges by intelligently distributing computation tasks among network nodes, leveraging the collective resources, and capabilities of participating devices. We will examine various strategies, algorithms, and frameworks that facilitate efficient and effective cooperative computation offloading, resulting in improved network performance, reduced latency, and optimized energy consumption.

4.4.1. State of the art

Most of the studies above only consider two options for the offloading decision: UEs either perform the computation locally or offload it to a UAV for edge computing. However, computation services can be performed using different distant surrogate devices (servers). In the following section, a third option for addressing the computation offloading problem is introduced. Besides the obvious choice of executing an offloaded task by itself, the UAVs in this subsection have two possible offloading choices to relieve their computing pressure: (i) they can send the collected tasks to a powerful nearby BS through a wireless local access network (WLAN), or (ii) they can send them to a more powerful cloud server. The possible cooperative manners are illustrated in Fig. 9. The studies investigating the UAV-enabled cooperative computation offloading system are compared in Table 13.

The UAV in [192] serves as a computing platform but also as a DF relay, and the computation bits generated by UEs can be computed locally, offloaded to the UAV for computing, or transmitted to the AP for processing via UAV relaying. Mei et al. in [193] studied a UAV-enabled edge-cloud system where multiple GUEs offload tasks to a UAV implemented with edge and cloud computing capacities. The EC of the UAV for flying, computing and sending computation results back to UEs was minimized through jointly optimizing bandwidth allocation, downlink transmitting power, computation capacity allocation, and the 3D trajectory and flight time of both rotary-wing and fixed-wing UAVs. Yu et al. in [194] proposed deploying UAV to facilitate MEC service provisioning to a set of stationary IoT devices that can optimally choose to offload partial computation tasks to the UAV or BSs. In [195], the UEs can directly offload computation tasks to UAVs or BS, or communicate with the BS via a UAV relay, depending on the channel conditions of UE-UAV, UE-BS and UAV-BS links, and the average latency of all UEs was minimized, including the transmission time of the uplink, computing time for the UAV or BS, and task queuing time at the UE.

A three-layer UEs-Edge-Cloud network structure was studied in [196], where UEs can offload their computation tasks to either UAVs or the cloud, depending on the size and delay tolerance of tasks. With the goal of minimizing UEs' weighted-sum of latency and ECs, a Q-learning based reinforcement learning algorithm was proposed in [196], where the objective function was defined as states in different time intervals during a timeslot, and each different offloading decision was defined as an action. By setting UEs' random mobility motion and UAVs' uniform circular motion in a fixed area, the three-layer architecture has the weighted time and energy consumptions saved for 5 percent, compared with the conventional two-layer architecture.

In [197], the UAV was enabled to offload computing requests collected from GUEs, to terrestrial edge servers or neighboring UAVs, while UEs and UAVs should pay for the allocated communication

Table 13
Comparison of literatures about UAV-enabled cooperative computation offloading.

Ref.	Platforms	Offloading type	Features	Conclusions
[192]	UAV+AP	Partial	• The UAV acts as a server and a DF relay; • The estimation of UEs mobility is adopted to design the proactive UAV trajectory;	• The UAV tends to forward computation bits to the AP as the increasement of tasks requirements; • The UAV flies closer to UEs with high demands for computing.
[193]	UAV+cloud	Full	• DL instead of UL optimization; • UE-UAV channels are modeled by the probability of LoS/NLoS path;	• UEs' ECs and latency are related with UEs' density; • Propulsion energy of UAV keeps stable for different densities of UEs.
[194]	UAV+APs	Full	• The UAV serves as a server and a decode-and-forward relay; • The EC of UAV receiving tasks from UEs is involved.	• The benefit of the cooperative between UAV and APs is verified.
[195]	UAVs+AP	Full	• UAVs serve as servers and relays; • UE-UAV channels are modeled by the probability of LoS/NLoS path; • Task queuing time at each UE is involved.	• Both the maximum and average UEs' latency decrease as the no. of UAVs increases.
[196]	UAVs+cloud	Binary	• Intelligent algorithm.	• The increase in task tolerant delay results in increased ratio of successful processed tasks.
[197]	UAVs+BS	Partial	• Time-varying UE-UAV channels are modeled as finite state Markov channels; • UEs generate random tasks in each timeslot.	• The centralized algorithm achieves more utility and lower task drop rate than that of distributed algorithm; • The proposed cooperative method achieves better performance than that of non-cooperative schemes.
[198]	UAVs+BSs +cloud	Binary	• Orthogonal frequencies are allocated for different UAVs and BSs; • Inactive/active status of BSs is adjustable.	• The proposed algorithm converges to a stable coalitional structure and the mixed-strategy NE of servers.
[199]	UAVs+APs	Binary	• The height-dependent pathloss channel model is adopted; • The convergence of proposed algorithm is analyzed.	• UEs' ECs increase with UEs computing capacity, and decrease with the no. of UAVs; • UEs' ECs decrease as the latency constraint becoming tighter.
[200]	UAV+AP	Partial	• UAV operates in a FDD model to perform task receiving from UEs, relaying to AP and results downloading from AP; • DLs of UEs are optimized.	• The proposed algorithm is more capable of handling computation-intensive and latency-critical tasks; • The UAV's trajectory is greatly affected by the relative location of AP and the distribution of UEs' task sizes.
[201]	UAV+idle UEs	Partial	• UAV serves as a server and a ET; • Idle UEs use harvested energy to process tasks from active UEs; • The received data at UAV can be computed in the subsequent timeslot.	• Trajectory optimization domains the minimization of the system EC; • UAV's acceleration optimization has great effect on UAVs' EC; • Active UEs tend to process equal volume of bits in each timeslot for energy saving.
[202]	UAV+BS+RSUs +vehicles	Full	• From a long-term perspective, the user association is investigated; • Vehicle mobility and content caching models are adopted for link reliability.	• The QoE is getting worse as the density of vehicles increases.
[203]	UAV+BSs	Binary	• The smooth-turn and boundary Gauss–Markov mobility models are applied to simulate the UAV trajectory and UEs movements, respectively;	• The proposed algorithm achieves better average utility performance.
[204]	UAVs+BSs	Binary	• Some BSs are connected to the cloud, while others are not connected; • UE-UAV channels are considered to be millimeter wave.	• The performance of the proposed algorithm is evaluated for different types of tasks.

and computation resources. Particularly, the offloading system was modeled as a semi-Markov decision process, considering random devices' requests and time-varying communication channels. The UAV's utility, defined as the difference between income and expenditure, was maximized by the state representation learning-based DRL algorithm proposed for the centralized system and the Q-value transfer-based DRL algorithm proposed for the distributed framework.

In [198], a two-level stochastic hierarchical game model, including a cooperative game and a set of noncooperative games, was proposed for UAV-enabled MEC networks, with two types of computing platforms: the terrestrial BS connected to cloud servers and UAVs with limited computing capacity. Besides, the UAVs were assumed to be inactive when they periodically returned to docking stations to perform battery recharge; otherwise, they were active for computing tasks offloaded from ground UEs. Moreover, considering the dynamics of processing resources, the mobility of UAVs, and the varying sizes of UEs' computing tasks, a Markov decision process was defined, and two algorithms for joint coalition formation and RL were proposed to minimize the long-term payoffs of BSs and UAVs.

The ground UEs can offload computation bits to flying UAVs or ground edge servers in [199], where the EC of UEs was minimized under the constraint of the UAV's energy budget. The UAV in [200] not only helps the UEs complete their computing tasks but also acts as a relay to further offload their tasks to the AP for computing, and the

UAV operates in a FDD mode with separate bandwidths allocated for task reception from UEs, task transmission to the AP, and computation results transmission to UEs. In [201], the active UEs can offload partial computation bits to the UAV or nearby idle UEs; meanwhile, the UEs can harvest energy from the UAV to prolong their lifetime. The UAV's energy consumption for computing, charging for UEs, flying and velocity acceleration was minimized under the energy budget of UEs, including consumed and conserved energy. Moreover, research results in [201] show that acceleration has a great effect on the required energy of UAV.

[202] investigated how to select the optimal platforms, i.e., neighboring vehicles, RSUs and UAV, to provide video chunks for a vehicular user (VUE). During a timeslot, the number of video chunks and links reliability achieved by each platform-to-VUE channel were theoretically derived, considering vehicle's mobility and heterogeneous caching resources. Then, the user association problem was formulated as a VUE's long-term utility (benefit for the QoE subtract handover cost) maximization, which was solved by an improved Dijkstra algorithm.

Considering the system dynamics caused by the flying UAV, UE mobility and time-varying task arrivals, [203] aimed to schedule UEs' offloading decisions during each timeslot, such that the expected long-term computation performance of UEs could be maximized. Specifically, an exponential utility function, related to UEs' EC and experienced delay, was used to quantify UE's computation experience, and the non-cooperative interactions among UEs were modeled as a stochastic game, following the developed DRL algorithm, which enables each MU to behave at an approximated Nash Equilibrium only with local information.

In [204], UAVs were implemented as a complementary form of ground BSs for providing UEs with computing resources, and BSs were equipped with MEC capacity or connected to the cloud. The goal was to maximize the number of UEs served by the network within their latency constraints, and this problem was formulated as a user association problem to decide whether to offload computing tasks and which platform to offload.

4.4.2. Open challenge

With recent developments, location-based services and augmented reality applications are being implemented extensively, particularly on mobile devices. For processing a growing volume of data, mobile devices are extremely resource-hungry. In an offloading scenario, competition among devices will continue to be a challenge, as each device will seek to obtain offloading services as quickly as possible. Due to this competition, an efficient resource scheduling algorithm is required. Furthermore, resource allocation must be maintained so that tasks with lower priority will not consume too many resources. Additionally, when the resource is freed after execution, it should be immediately available for another task. This is even more challenging when both ground BSs and UAVs are involved, and the task generation for users is stochastic. A dynamic resource allocation scheme is also necessary for cooperative UAV-MEC networks, which needs further study.

4.5. Computation offloading from UAV

UAV swarms are widely applied for service provisioning in many domains, such as package delivery, farming, and network connectivity. However, the UAV swarm faces inherent physical constraints as all computation and storage are conducted on board, and its advancement is restricted by its limited computational, storage, and communication capacities. Due to their poor onboard resources, UAV swarms cannot effectively perform resource-intensive applications, which usually have the following characteristics:

- Big data volume: Sensors or cameras on UAV swarms generate big streams of data traffic, which requires huge communication resources [205].
- Huge demand for processing power: Applications such as object recognition, navigation, and AI processing require intensive processing power to accomplish.

On the other hand, real-time applications have a very strict latency requirement, but the UAV individual or the ad hoc virtual cloud still have limited computational resources, which produce long data processing times and cannot meet the latency requirement of real-time applications. Fortunately, the edge servers deployed close to the UAV swarms, brought together by the ad hoc virtual cloud, are able to provide computational resources with lower delay, less network pressure, and more efficient data processing for real-time applications.

4.5.1. State of the art

Recently, many works have focused on enhancing the applications of UAVs, especially in the perspective of computation offloading. We list the state-of-art litterateurs in Table 14.

The authors in [206] provided two offloading methods: air-offloading and ground-offloading. If local computing resources are insufficient, the UAV can offload its computation tasks to nearby UAVs, called air-offloading. Furthermore, if air-offloading fails, the UAV should offload tasks to micro- or mini-clouds co-located with BSs, and this method refers to ground-offloading. Specifically, the UAV should offload its computation bits to a nearby UAV with the minimum response latency, and if the ground-offloading is active, the edge server providing the minimum response time will be selected.

Liu et al. in [207] studied a UAV-Edge-Cloud computing system, where the UAV with limited onboard resources offloads partial computation bits to edge servers or the cloud for processing, with the help of neighboring UAVs. Through jointly optimizing the workflow assignment and multi-hop routing scheduling, both computing and routing costs were minimized. A Markov approximation-based algorithm was firstly proposed to obtain near-optimal solutions for the formulated NP-hard non-convex problem, and then an online algorithm combining Markov approximation and Lyapunov optimization was developed for highly dynamic environments.

Messous et al. in [208] studied the computation offloading for a set of capacity-limited UAVs, which have to offload intensive computation tasks to a nearby BS via WiFi or a powerful edge server via cellular links. They formulated the offloading problem as a non-cooperative theoretical game with UAVs as players and three pure strategies referring to local computing, offloading to the BS, and edge server. The existence of a Nash Equilibrium was proved, indicating that the best possible tradeoff between energy-consumption, delay and computation cost can be achieved.

The authors in [212] considered offloading a fraction of machine learning tasks to a MEC server to relieve the pressure on UAV for target tracking. A hierarchical ML task distribution framework was proposed, where the UAV and MEC server were respectively embedded with lower and higher layers of the convolutional neural network (CNN) model. Considering computing and communications, the ML task processing delay was minimized through optimizing offloading decisions while ensuring data quality.

Focusing on both security and efficiency, the UAV offloading problem was investigated in [210,211], where the UAV offloads task bits to the BS/AP, in the presence of a ground passive Eve. In [211], the UAV is associated with multiple ground BSs, and the total energy-consumption for local computation, communication transmission and flight-propulsion was minimized via jointly optimizing UAV trajectory, computation task allocation, BSs selection and transmit power allocation. In [210], a hovering UAV tries to offload computation tasks to an AP co-located with an edge server, and the AP also imposes jamming noise on the Eve to combat eavesdropping. Two energy-efficient computation offloading problems, subject to both time-duration and security constraints, were respectively formulated for the UAV-MEC systems with a passive eavesdropper at a fixed/random location.

Table 14
Comparison of literatures about UAV offloading tasks.

Classification	Ref.	Design objective	Optimization items	Computing methods
Multi-UAV	[206]	Energy and latency efficiency	Computing platform selection	Local computing, offloading to a BS or an edge server
	[207]	Computation and routing cost	Workflow assignment and routing scheduling	Offloading to multiple edge servers and cloud servers
	[208]	Tradeoff between energy consumption, delay and computation cost	Offloading platform selection	Local computing, offloading to a BS or an edge server
	[209]	Latency	Offloading platform selection	Local computing, offloading to nearby UAVs or a cloud server
Single UAV	[210]	Energy	Transmit power and computation bits allocation, offloading time optimization	Local computing, partially offloading to an edge server
	[211]	Energy	Offloading platform selection, transmit power and computation bits allocation, UAV-2D trajectory and velocity optimization	Local computing, partially offloading to multiple BSs
	[212]	Latency	Computation bits allocation, inference error and the weight of task failing penalty optimization	Local computing, partially offloading to an edge server

According to the variety of service requirements, UAVs in [209] were classified into two types: cloud service support (CSS) UAVs and local service support (LSS) UAVs. CCS UAVs can either offload tasks to other CCS UAVs or the cloud, and the closed-form of the offloading decision was derived, through analyzing the response delay by using queuing theory. For LSS UAVs, the average age of information control algorithm for the status update system was designed to reduce the average age of information and guarantee the timeliness of information.

4.5.2. Open challenge

In the UAV-MEC system, selecting the decision-making process is an open problem. Specifically, the decision-making process can be either centralized or decentralized. A central controller is responsible for making all decisions in a centralized system. However, in this technique, the controller may face network congestion due to the unequal transmission power of UAVs and the varying wireless transmission characteristics. The decentralized technique can be a viable solution because each UAV uses the information available from other UAVs and thus decides whether to offload or not. Communication and computational aspects must be jointly considered for dealing with such a scenario.

5. Future research directions

As shown in the previous sections, the applications of UAVs have attracted a lot of attention in recent years due to their ability to provide on-demand radio connections for UEs while, at the same time, enabling LoS communications because of their ability to migrate obstacles and interference. Despite the fact that the UAV-enabled network is a complex cyber–physical system, there are many practical constraints as well as the induced issues that are worthy of future investigation. This section discusses some of these promising topics to inspire future works.

5.1. Channel modeling

The modeling of wireless channels for UAV-to-ground/ground-to-UAV links is an active and ongoing research area for many research groups. While the small-scale fading can be mainly modeled as classical Rayleigh, Rician or Nakagami-m fading models, the large-scale pathloss and shadowing modeling is usually more involved. Most existing models can be classified into three categories: the free-space LoS model, the modeling based on altitude- and angle-dependent channel parameters, and the probabilistic LoS channel modeling. Specifically, [123] adopted a distributed model, the predictive control framework, and the minimum spanning tree (MST) for improving the network connectivity performance. [152] adopted the Rayleigh product channel, which is a product of a series of independent and identically distributed (i.i.d.) Rayleigh channels, and [165] adopted the segmented ray-tracing channel model, which can distinguish different propagation scenarios caused by different locations of the UAV and users.

However, UAV-aided wireless networks demand techniques to predict channel variations introduced by their mobility, and this challenge is magnified when source and destination nodes are also mobile. In addition to the trajectory, the UAV's heading angle to quantify link performance has to be controlled to ensure optimal performance of the ground-to-UAV links. Furthermore, the Doppler effect and multipath fading in air-to-air communications need to be characterized while considering UAVs' altitude as well as antennas' movement. Therefore, to establish and maintain link connectivity, advanced A2G channel models and communication protocols that allow robust, reliable, and adaptive response to dynamic environments should be developed.

5.2. Interference avoidance and management

On the one hand, LoS links of air–ground transmissions can improve the communication performance of GUEs, but they may also increase the interference on other transmission links. When multiple nodes send data to one UAV simultaneously, they can cause serious mutual interference and transmission conflict. It is necessary to tackle the interference avoidance problem to improve the utilization rate of UAVs. Among them, beamforming and tracking technology based on the trajectory prediction of UAVs are considered to have good research potential. The cooperation mechanism is also a potential way to reduce the interference of ground devices and improve transmission efficiency. Considering that the mobility of UAVs is limited by tasks and cannot cover all the required nodes, the ground devices can cooperate with each other to form transmission clusters. The node in the dominant position can be the cluster head to transmit data with UAVs.

Moreover, one prominent interference mitigation tool is Coordinated Multi-Point (CoMP) transmission and reception (and its variants). The new challenge here is that aerial UEs receive interfering signals from more ground BSs in the downlink, and their uplink signals are visible to more cells due to better LoS propagation conditions. Thus, the methods need to scale for a large set of cells without much complication due to the requirements on additional pilots, synchronization, scheduling, and so on. The optimal grouping strategy for cooperation among BSs, the tradeoff between overhead, coordination complexity, and interference mitigation gain, are still open problems. Interference

can also be handled by receiver techniques such as interference rejection combining and network-assisted interference cancellation and suppression [213]. Comparing the sizes of small UAVs to the sizes of smart hones, it is more feasible to equip small UAVs with more antennas, which can be used to cancel or suppress the interfering signals from more ground BSs. With multiple antennas, beamforming that enables directional signal transmission or reception to achieve spatial selectivity is also an effective interference mitigation technique.

5.3. Performance evaluation

One of the major challenges in the development of the UAV network is determining appropriate methods and metrics to evaluate the performance of different components. The evaluation of the communication, computation, and control of UAVs, as well as their physical processes, can be performed with a variety of performance indicators and testing platforms. Yet there are still no theoretical foundations or practice guidelines for evaluating the cyber–physical characteristics of UAV-enabled networks. Moreover, the heterogeneity of multiple subsystems and numerous features of the cyber/physical domain, hardware, and software are all included in the cyber–physical model of UAV networks. All of these could lead to state space extension, which complicates system verification and substantially raises verification costs [214]. On the one hand, cross-layer resource scheduling requires considerable research and development. High heterogeneity exists in cross-layer resources, even in the same layer, bringing the challenge of resource scheduling. Thus, it is impossible to verify the coupling depth between each component. What is worse, the contributions of the coupling to system improvement are also difficult to quantify.

Moreover, due to the strict latency and energy requirements, some mechanisms for QoS guarantees based on the UAV-Edge-Cloud model are further required. Application partitioning for efficiency in computation offloading is a complex problem since identifying the resource-intensive components in workloads remains an issue. Besides, the stochastic characteristics of wireless networks could cause unpredictable latency and energy consumption [215], and QoS-aware communications are critical to fully realizing the power of the UAV-Edge-Cloud model. Therefore, much effort, such as using 5G techniques, must be made to address the communication issues associated with real-time processing.

5.4. Security and privacy

The closed loop of UAV-enabled networks contains multiple components and cross-layer resources (i.e., sensing, communication, computing and control), which makes it open to outside attacks since security issues in any one component will leave the entire system dysfunctional. Under the terrestrial and aerial layers, the attacks could be performed from either the physical or the cyber domain. The former refers to the attacks that specifically tamper with the physical components of UAV-enabled networks, such as charging batteries or hiding goals. The latter implies attacks that are carried out via metalware and software. This can be done by jamming or spoofing UAV communications or by fabricating sensor data. In conclusion, communication security, control security, feedback security, sensing security, and storage security are among the primary security requirements for UAV networks.

Along with security challenges, there are privacy issues. Regardless of hierarchy, the expanding deployment of UAV networks generates additional perceived data, such as geographic addresses and some personally identifiable and sensitive data, for example, when carrying out a surveillance operation. Any individual can be collectively profiled using such information. Thus, it is horrifyingly possible to plot a physical attack against buildings, people, or installations when breaking into the storage, communication, or actuation modules of UAV networks. The cloud service platform can be used to mine various pieces of information from multiple UAVs with varying mission specifics, especially when the cloud is involved. Therefore, when developing security solutions for UAV networks, privacy leakage should always be taken into account. The authors in [216] investigate the security and privacy requirements of the Internet of UAVs and present possible solutions for challenging problems such as privacy leakage, data confidentiality protection, and flexible accessibility. The authors in [217] have reviewed the key security, privacy, and safety issues related to the use of civilian drones in the national airspace. They explore the security requirements imposed by such systems' essential operating environments while identifying the physical and digital risks that these systems face. And yet sharing this data in a secure and effective manner tends to be challenging.

5.5. Machine learning for resources allocation and flight control

The adoption of UAVs in various communication-based applications is expected to improve coverage and spectral efficiency, compared to traditional ground-based solutions. However, this new degree of freedom that will be included in the network will also add new challenges, including the plethora of configuration parameters that are making UAV-aided next-generation communications exceedingly complex. Specifically, the dependencies between parameters and their joint optimization usually have a nonlinear impact on network performance. The complexity is further increased in the case of dense deployments and coexistence with shared resources. While classical optimization approaches fail in such conditions, machine learning is able to handle complexity. So far, ML-based techniques have been explored for a variety of problems in networking [218–221]. Successful solutions are applied to fields ranging from configuring physical layer parameters to traffic prediction.

The latest advancements in ML techniques have opened up new opportunities for the UAV-enabled systems and have led to the possibility of realizing highly autonomous UAV operations while enhancing the network performance under complex and dynamic scenarios. For example, in a UAV-assisted sensor network, a deep reinforcement learning-based flight resource allocation framework is proposed in [222] and a deep deterministic policy gradient-based maneuver control scheme is developed in [223], to minimize the data packet loss during data collection. UAVs can dynamically adapt their positions, flight paths, and motion control to support GUEs, by using various reinforcement learning algorithms [224]. Additionally, one may efficiently deploy and control UAVs and predict GUEs' behaviors by applying neural network techniques and data analysis. The radio environment maps and 3D channel models can also be effectively learned by leveraging machine learning techniques [225]. These radio environment maps can then be leveraged to deploy and operate UAV communication systems to their full potential. However, developing ML solutions with a fast learning speed is of great importance. In this regard, hierarchical learning principles allow improving learning speed by decomposing complex joint optimization problems into multiple sub-problems [226]. In addition, it is important to jointly design the flight control of the UAV and its data aggregation scheduling by taking the battery levels and propulsion energy consumption into account, since different patrol velocities of the drone can incur different energy consumptions of the flight control.

5.6. AI techniques and intelligent algorithm

The challenges of end node management, real-time decision making, fast user association and resource allocation fall into the research of artificial intelligence (AI), which is considered a promising technique to address such problems through adaptive modeling and intelligent learning. Different from traditional optimization methods (e.g., convex optimization, dynamic programming and game theory), an AI-based solution has continuous learning ability for the dynamic environment and can make real-time inferences with low computational complexity.

Specifically, due to the 3C resources available in UAV-enabled wireless systems, AI-based models can train on cloud platforms with varying accuracy and abilities, depending on the available resources. Some complex models can train in places with more resources, e.g., the central cloud, and then downloaded to places with fewer resources, e.g., UAVs. Additionally, inference can make in edge nodes, depending on the latency requirements of the applications or users.

Some researchers have given insights into the application of AI techniques to enhance the performance of UAV-aided communications. For example, [227] proposed a distributed DNN-based offloading strategy to minimize the weighted sum of delay and energy consumption of UEs, in which the offloading decisions were used to train three separate DNNs, of which the one with the least training loss was selected to obtain near-optimal task offloading decisions. [228] has brought AI into the UAV's data offloading process for a multi-UAV and multi-server MEC system. Specifically, based on the theory of the stochastic learning automata, the autonomous MEC server selection was performed by the UAVs, and AI was embodied for each UAV to determine the amount of data to be offloaded to the selected MEC server. A non-cooperative game was also formulated with the objective of maximizing each UAV's utility, which proves the existence of at least one NE. To autonomously approach the NE point, a best response dynamics framework and two alternative reinforcement learning algorithms were developed, and their tradeoffs in terms of efficiency and effectiveness were discussed. In [229], the AI-based joint resource scheduling framework with two different AI-based mechanisms, i.e., DNN-based and DRL-based architectures, was proposed for a heterogeneous MEC system. They concluded that the DNN-based solution with online incremental learning applies the global optimizer and therefore has better performance than the DRL-based architecture with online policy updating but requires more training time. Aside from the several potential uses of AI techniques for various applications of UAVs, AI can be utilized for intelligent collision avoidance in multi-UAV deployments that crowded with flying UAVs. Enhancing the application of AI for massive MIMO channel estimation, white space identification, and intelligent energy management could be another future option.

5.7. Space-air-ground integrated networks

Since satellites, UAVs, and BSs can complement each other, their integration, namely the space-air-ground integrated network (SAGIN), is proposed as a promising next-generation wireless network to serve the massive IoT devices with delay-oriented service requirements [1,230]. The overall objective of SAGIN is to provide user devices with improved and flexible end-to-end services, which are categorized into three main segments: space-based, air-based, and ground-based layers. These three segments can work independently or cooperatively, and by integrating heterogeneous networks among the three segments, it is easy to build a hierarchical broadband wireless network. UAVs are then deployed to create a multi-tier UAV network, and the ground network is made up of mobile users, vehicles, and radio access infrastructures. Although SAGIN is becoming ever more important for B5G wireless communications, designing the integrated network is a challenging task that involves channel modeling [231], optimal deployment, energy-efficiency, latency, traffic offloading [232,233], management of HetNets, network security [234], etc.

Due to its unique qualities, such as heterogeneity, self-organization, and time-variability, SAGIN not only provides considerable benefits for a variety of practical services and applications, but it also faces numerous previously encountered obstacles. The restricted and imbalanced network resources in all three network segments of SAGIN, in contrast to conventional ground or satellite networks, make it challenging to achieve the greatest traffic delivery performances. System integration, protocol optimization, resource management, and allocation in SAGIN are therefore of great significance.

6. Conclusions

In this article, the performance gains and potential applications of UAVs in wireless networks are first introduced. Then, we survey the roles of UAVs in wireless networks, such as aerial BSs, relay nodes, and edge computing servers, respectively. Specifically, UAVs as aerial BSs for data collection, energy provision and information broadcasting are identified. The implementation issues and state-of-the-art works related to UAV relaying are also discussed in terms of relaying schemes, UAV trajectory/placement design, security and reliability issues, and resources optimization for the UAV relaying system. As for UAVs working as edge computing servers, the implementation issues are first outlined, and then we comprehensively review the computation offloading from terrestrial UEs to a single UAV, multi-UAV and multi-platform. Computation offloading from UAVs is also being researched. Finally, some interesting and emerging research trends that may contribute to the improvement of UAV-assisted wireless networks are discussed.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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